

Importance-weight guidance under left-to-right block diffusion

Experiment record for the ICLR submission. Companion to `RESULTS_BD_V4.pdf`, which remains the record of the full v4 round; this document covers only what the submission needs: (i) an audit of the like-for-like conditions behind the left-to-right (l2r) experiment, (ii) our method under l2r, (iii) the published D-CBG baseline under the *same* sampler, (iv) the mechanism by which block diffusion outperforms autoregressive decoding once guidance is applied, and (v) the experiments still to run before submission. Family 0.05, scenario `except_0`, 1,000 held-out OD pairs, RAW (no post-processing) throughout. Parenthesised grey values are the corresponding first-hitting (random reveal order) numbers from the v4 report.

1. What changed, and what did not

The mask kernel commits one masked position per denoising step. Two rules for choosing that position:

- `first_hit` — a uniformly random still-masked position (the default used throughout the v4 report).
- `l2r` — always the left-most still-masked position.

The first-hitting *time* schedule ($t \leftarrow t \cup^{1/\#\text{masked}}$) is identical in both, so the ablation isolates the reveal **position** rule and nothing else. Both rules live in a single branch on each side — `bd_models.py::_denoise_block_mask_guided._pick` for our method and `dcbg_plugin.py::plan_dcbg_mask` for the baseline — and the baseline's branch is a verbatim copy of ours, so the two methods are compared under exactly the same sampler. No guidance formula was touched: D-CBG still computes $\log p_\theta + \gamma \cdot \log p_\phi$ at the reveal position, ours still computes self-normalised weights $\exp(\gamma \ell)$ over $n_{\text{is}} = 100$ block candidates.

2. Fairness audit of the l2r experiment

2.1 Invocation parity

The two runs were launched by the same script body; the commands differ in one token.

held fixed	value
backbone	<code>BD_porto_v3_normal_mask_blk{B}_v4_bd.pth</code>
discriminator	<code>BDdisc_f0.05_p1_{e0,e99}_model_blk{B}_v4.pth</code>
OD pairs	the 1,000 reserved <code>except_0</code> rows, <code>RandomState(777).permutation(...)[:1000]</code>
importance samples	$n_{\text{is}} = 100, \gamma = 1.0$
batch / seed	100 / 7, re-seeded before each of base, IW, adj+IW
scoring	RAW, <code>evaluate_em_pc</code> , arrival recomputed from the generated endpoint
differs	<code>-order l2r</code>

2.2 Verified, not assumed

- Identical OD pairs:** in both runs and in all three configurations, the generated path's first vertex equals the reserved pair's origin for **1,000 / 1,000** rows.
- The order flag takes effect, and only where it can:** at blk4 the l2r and first-hitting runs return different paths for 68% of pairs under adj+IW (316/1000 identical), 30% under base. At blk1 — where a block holds one position and both rules must commit the *same* position — the aggregate metrics agree to ± 0.002 on $v_{\wedge a}$ (0.542 vs 0.540) even though 53% of individual paths differ.
- Where the blk1 divergence comes from, and why it is useful:** `first_hit` spends a `torch.multinomial` draw on the position choice and `l2r` does not, so the RNG streams separate from the first step. The blk1 row is therefore a free noise-floor measurement at this eval size: $v_{\wedge a} + 0.012$ (base), $+0.007$ (IW), $+0.002$ (adj+IW). Read every other row against ≈ 0.012 .
- Both regimes measured:** seen (discriminator trained on `except_0`) and unseen (trained on `except 1–99`, evaluated zero-shot on `except_0`), so no conclusion rests on the discriminator having seen the test scenario.

2.3 One residual mismatch, disclosed

`gen_bd_uncond_pool.py` generates the unconditional negative pools with `model.plan(None, None, n_samples=...)` and no order argument, i.e. under **first-hitting**. Those pools are the model-negatives for both our discriminators and D-CBG's classifiers. So when we evaluate under l2r, the reference distribution p_{ref} the scorer was trained against is not the p_ϕ being guided. Consequences:

- The **IW vs D-CBG comparison remains fair** — both scorers carry the identical handicap.
- The **absolute** l2r numbers for both methods are pessimistic by an unknown amount: a scorer trained on l2r negatives would face a harder, better-matched negative class.
- Fixing it is cheap and is listed as experiment E2 in §6.

3. Our method under l2r

3.1 Seen (discriminator trained on except_0)

blk	base valid	base arr	base v _Λ a	base EM	base PC	adj+IW valid	adj+IW arr	adj+IW v _Λ a	adj+IW EM	adj+IW PC
1	0.312 (0.302)	0.996 (0.994)	0.312 (0.300)	0.275 (0.271)	0.289 (0.283)	0.925 (0.928)	0.573 (0.567)	0.542 (0.540)	0.361 (0.374)	0.415 (0.425)
2	0.268 (0.251)	0.963 (0.949)	0.262 (0.238)	0.244 (0.233)	0.255 (0.240)	0.867 (0.492)	0.678 (0.846)	0.577 (0.401)	0.440 (0.388)	0.515 (0.423)
4	0.293 (0.279)	0.976 (0.932)	0.285 (0.262)	0.262 (0.243)	0.275 (0.258)	0.765 (0.389)	0.732 (0.873)	0.556 (0.328)	0.434 (0.334)	0.503 (0.357)
8	0.272 (0.229)	0.964 (0.938)	0.256 (0.214)	0.232 (0.202)	0.248 (0.214)	0.760 (0.331)	0.685 (0.862)	0.487 (0.273)	0.392 (0.295)	0.480 (0.309)
16	0.259 (0.206)	0.930 (0.923)	0.240 (0.194)	0.221 (0.180)	0.238 (0.190)	0.747 (0.298)	0.653 (0.865)	0.447 (0.250)	0.389 (0.230)	0.485 (0.255)
32	0.251 (0.185)	0.898 (0.870)	0.221 (0.160)	0.202 (0.157)	0.221 (0.168)	0.723 (0.265)	0.635 (0.825)	0.427 (0.215)	0.376 (0.219)	0.468 (0.237)
64	0.235 (0.141)	0.901 (0.827)	0.202 (0.119)	0.178 (0.129)	0.199 (0.134)	0.828 (0.229)	0.511 (0.795)	0.393 (0.162)	0.359 (0.176)	0.474 (0.198)

3.2 IW only, no adjacency masking — seen

blk	base valid	base arr	base v _Λ a	base EM	base PC	IW valid	IW arr	IW v _Λ a	IW EM	IW PC
1	0.312 (0.302)	0.996 (0.994)	0.312 (0.300)	0.275 (0.271)	0.289 (0.283)	0.331 (0.324)	0.992 (0.991)	0.328 (0.321)	0.287 (0.283)	0.303 (0.298)
2	0.268 (0.251)	0.963 (0.949)	0.262 (0.238)	0.244 (0.233)	0.255 (0.240)	0.322 (0.328)	0.963 (0.951)	0.317 (0.312)	0.292 (0.306)	0.305 (0.315)
4	0.293 (0.279)	0.976 (0.932)	0.285 (0.262)	0.262 (0.243)	0.275 (0.258)	0.337 (0.332)	0.965 (0.936)	0.320 (0.304)	0.305 (0.295)	0.318 (0.310)
8	0.272 (0.229)	0.964 (0.938)	0.256 (0.214)	0.232 (0.202)	0.248 (0.214)	0.321 (0.294)	0.963 (0.923)	0.303 (0.254)	0.289 (0.270)	0.304 (0.279)
16	0.259 (0.206)	0.930 (0.923)	0.240 (0.194)	0.221 (0.180)	0.238 (0.190)	0.312 (0.260)	0.934 (0.917)	0.281 (0.236)	0.259 (0.211)	0.280 (0.230)
32	0.251 (0.185)	0.898 (0.870)	0.221 (0.160)	0.202 (0.157)	0.221 (0.168)	0.322 (0.244)	0.894 (0.867)	0.267 (0.207)	0.259 (0.204)	0.283 (0.220)
64	0.235 (0.141)	0.901 (0.827)	0.202 (0.119)	0.178 (0.129)	0.199 (0.134)	0.307 (0.201)	0.879 (0.854)	0.247 (0.157)	0.251 (0.160)	0.271 (0.177)

3.3 Unseen (discriminator = except 1–99 → except_0, zero-shot)

blk	base valid	base arr	base v _Λ a	base EM	base PC	adj+IW valid	adj+IW arr	adj+IW v _Λ a	adj+IW EM	adj+IW PC
1	0.312 (0.302)	0.996 (0.994)	0.312 (0.300)	0.275 (0.271)	0.289 (0.283)	0.928 (0.925)	0.556 (0.558)	0.527 (0.530)	0.354 (0.362)	0.406 (0.412)
2	0.268 (0.251)	0.963 (0.949)	0.262 (0.238)	0.244 (0.233)	0.255 (0.240)	0.863 (0.488)	0.677 (0.848)	0.574 (0.415)	0.431 (0.367)	0.507 (0.408)
4	0.293 (0.279)	0.976 (0.932)	0.285 (0.262)	0.262 (0.243)	0.275 (0.258)	0.765 (0.390)	0.738 (0.875)	0.561 (0.331)	0.439 (0.335)	0.506 (0.357)
8	0.272 (0.229)	0.964 (0.938)	0.256 (0.214)	0.232 (0.202)	0.248 (0.214)	0.758 (0.331)	0.690 (0.863)	0.490 (0.273)	0.393 (0.295)	0.481 (0.308)
16	0.259 (0.206)	0.930 (0.923)	0.240 (0.194)	0.221 (0.180)	0.238 (0.190)	0.748 (0.300)	0.653 (0.868)	0.448 (0.254)	0.389 (0.237)	0.485 (0.261)
32	0.251 (0.185)	0.898 (0.870)	0.221 (0.160)	0.202 (0.157)	0.221 (0.168)	0.723 (0.268)	0.633 (0.828)	0.425 (0.219)	0.378 (0.222)	0.469 (0.240)
64	0.235 (0.141)	0.901 (0.827)	0.202 (0.119)	0.178 (0.129)	0.199 (0.134)	0.828 (0.224)	0.510 (0.798)	0.392 (0.162)	0.359 (0.178)	0.474 (0.197)

3.4 IW only, no adjacency masking — unseen

blk	base valid	base arr	base v _Λ a	base EM	base PC	IW valid	IW arr	IW v _Λ a	IW EM	IW PC
1	0.312 (0.302)	0.996 (0.994)	0.312 (0.300)	0.275 (0.271)	0.289 (0.283)	0.327 (0.321)	0.992 (0.992)	0.324 (0.319)	0.283 (0.281)	0.299 (0.296)
2	0.268 (0.251)	0.963 (0.949)	0.262 (0.238)	0.244 (0.233)	0.255 (0.240)	0.328 (0.330)	0.966 (0.951)	0.321 (0.316)	0.298 (0.305)	0.311 (0.316)
4	0.293 (0.279)	0.976 (0.932)	0.285 (0.262)	0.262 (0.243)	0.275 (0.258)	0.336 (0.331)	0.972 (0.935)	0.326 (0.302)	0.303 (0.295)	0.316 (0.310)
8	0.272 (0.229)	0.964 (0.938)	0.256 (0.214)	0.232 (0.202)	0.248 (0.214)	0.323 (0.298)	0.963 (0.923)	0.304 (0.258)	0.290 (0.275)	0.305 (0.284)
16	0.259 (0.206)	0.930 (0.923)	0.240 (0.194)	0.221 (0.180)	0.238 (0.190)	0.315 (0.264)	0.934 (0.919)	0.283 (0.242)	0.263 (0.215)	0.284 (0.234)
32	0.251 (0.185)	0.898 (0.870)	0.221 (0.160)	0.202 (0.157)	0.221 (0.168)	0.322 (0.246)	0.896 (0.869)	0.269 (0.209)	0.263 (0.208)	0.285 (0.223)
64	0.235 (0.141)	0.901 (0.827)	0.202 (0.119)	0.178 (0.129)	0.199 (0.134)	0.306 (0.199)	0.879 (0.857)	0.248 (0.157)	0.250 (0.161)	0.270 (0.177)

- **I2r wins at every block size ≥ 2 , in every configuration.** base and IW-only gains grow monotonically with block size (base +0.024 at blk2 → +0.083 at blk64); adj+IW gains are large and roughly flat above blk2 (+0.176 ... +0.231). At blk64 adj+IW v_Λa goes 0.162 → 0.393.
- **A validity/arrival trade that comes out well ahead:** validity roughly doubles (blk4 0.389 → 0.765; blk64 0.229 → 0.828) while arrival falls (0.873 → 0.732; 0.795 → 0.511), and the product still rises by +0.228 / +0.231. EM and PC, which the trade could have damaged, rise as well (+0.183 EM, +0.276 PC at blk64).
- **Zero-shot transfer is unaffected:** the largest seen-vs-unseen difference anywhere in the I2r tables is 0.017, matching the v4 report's §4.2.
- **Block diffusion overtakes AR under guidance.** blk1 is a block-size-1 block diffusion model, i.e. left-to-right autoregressive decoding, and it is where I2r and first-hitting coincide. Against it, adj+IW at blk2 gains v_Λa +0.035, EM +0.079, PC +0.100; PC of every block ≥ 2 exceeds blk1's. The mechanism is §5.
- **Cost: none.** I2r removes one `multinomial` draw per step and changes no scorer-call count.

3.5 Family 0.1 — twice the removed edges

Edge-remove ratio 0.1 doubles the removed edges and roughly halves base validity, so it is the harder of the two worlds and the more discriminating test. Same protocol; parentheses are the family 0.1 first-hitting numbers from the v4 report §6.

Seen

blk	base valid	base arr	base v _Λ a	base EM	base PC	adj+IW valid	adj+IW arr	adj+IW v _Λ a	adj+IW EM	adj+IW PC
1	0.112 (0.115)	0.993 (0.993)	0.111 (0.114)	0.103 (0.108)	0.106 (0.110)	0.918 (0.927)	0.246 (0.250)	0.229 (0.234)	0.147 (0.144)	0.180 (0.177)
2	0.090 (0.096)	0.974 (0.960)	0.089 (0.092)	0.084 (0.089)	0.086 (0.092)	0.762 (0.245)	0.382 (0.717)	0.281 (0.163)	0.194 (0.174)	0.257 (0.200)
4	0.109 (0.089)	0.974 (0.949)	0.103 (0.084)	0.095 (0.075)	0.100 (0.081)	0.610 (0.169)	0.410 (0.841)	0.227 (0.137)	0.174 (0.141)	0.219 (0.152)
8	0.093 (0.077)	0.965 (0.933)	0.086 (0.071)	0.089 (0.073)	0.090 (0.074)	0.578 (0.143)	0.422 (0.804)	0.199 (0.101)	0.198 (0.123)	0.263 (0.131)
16	0.091 (0.075)	0.931 (0.934)	0.081 (0.069)	0.078 (0.067)	0.084 (0.071)	0.658 (0.132)	0.364 (0.819)	0.187 (0.091)	0.202 (0.103)	0.298 (0.115)
32	0.080 (0.072)	0.886 (0.884)	0.066 (0.061)	0.056 (0.062)	0.066 (0.066)	0.539 (0.118)	0.411 (0.793)	0.180 (0.076)	0.174 (0.088)	0.248 (0.099)
64	0.076 (0.040)	0.900 (0.865)	0.062 (0.032)	0.057 (0.034)	0.065 (0.036)	0.774 (0.119)	0.282 (0.744)	0.173 (0.068)	0.173 (0.094)	0.287 (0.105)

Unseen (except 1–99 → except₀)

blk	base valid	base arr	base v _Λ a	base EM	base PC	adj+IW valid	adj+IW arr	adj+IW v _Λ a	adj+IW EM	adj+IW PC
1	0.112 (0.115)	0.993 (0.993)	0.111 (0.114)	0.103 (0.108)	0.106 (0.110)	0.920 (0.932)	0.244 (0.248)	0.228 (0.232)	0.148 (0.145)	0.181 (0.177)
2	0.090 (0.096)	0.974 (0.960)	0.089 (0.092)	0.084 (0.089)	0.086 (0.092)	0.764 (0.243)	0.377 (0.720)	0.280 (0.164)	0.190 (0.174)	0.253 (0.198)
4	0.109 (0.089)	0.974 (0.949)	0.103 (0.084)	0.095 (0.075)	0.100 (0.081)	0.618 (0.168)	0.405 (0.841)	0.226 (0.136)	0.169 (0.141)	0.216 (0.151)
8	0.093 (0.077)	0.965 (0.933)	0.086 (0.071)	0.089 (0.073)	0.090 (0.074)	0.579 (0.145)	0.422 (0.803)	0.198 (0.101)	0.193 (0.124)	0.262 (0.132)
16	0.091 (0.075)	0.931 (0.934)	0.081 (0.069)	0.078 (0.067)	0.084 (0.071)	0.653 (0.134)	0.361 (0.819)	0.181 (0.092)	0.198 (0.102)	0.297 (0.115)
32	0.080 (0.072)	0.886 (0.884)	0.066 (0.061)	0.056 (0.062)	0.066 (0.066)	0.540 (0.117)	0.408 (0.800)	0.182 (0.079)	0.170 (0.084)	0.245 (0.097)
64	0.076 (0.040)	0.900 (0.865)	0.062 (0.032)	0.057 (0.034)	0.065 (0.036)	0.774 (0.121)	0.282 (0.745)	0.174 (0.070)	0.171 (0.095)	0.286 (0.105)

IW only, seen

blk	base valid	base arr	base v _Λ a	base EM	base PC	IW valid	IW arr	IW v _Λ a	IW EM	IW PC
1	0.112 (0.115)	0.993 (0.993)	0.111 (0.114)	0.103 (0.108)	0.106 (0.110)	0.127 (0.122)	0.997 (0.994)	0.126 (0.119)	0.110 (0.103)	0.116 (0.109)
2	0.090 (0.096)	0.974 (0.960)	0.089 (0.092)	0.084 (0.089)	0.086 (0.092)	0.121 (0.125)	0.967 (0.960)	0.117 (0.114)	0.110 (0.114)	0.114 (0.119)
4	0.109 (0.089)	0.974 (0.949)	0.103 (0.084)	0.095 (0.075)	0.100 (0.081)	0.116 (0.118)	0.979 (0.933)	0.112 (0.108)	0.103 (0.104)	0.108 (0.109)
8	0.093 (0.077)	0.965 (0.933)	0.086 (0.071)	0.089 (0.073)	0.090 (0.074)	0.122 (0.115)	0.961 (0.926)	0.111 (0.094)	0.111 (0.102)	0.116 (0.107)
16	0.091 (0.075)	0.931 (0.934)	0.081 (0.069)	0.078 (0.067)	0.084 (0.071)	0.113 (0.101)	0.928 (0.924)	0.095 (0.082)	0.097 (0.086)	0.105 (0.092)
32	0.080 (0.072)	0.886 (0.884)	0.066 (0.061)	0.056 (0.062)	0.066 (0.066)	0.112 (0.094)	0.897 (0.889)	0.091 (0.069)	0.097 (0.073)	0.103 (0.081)
64	0.076 (0.040)	0.900 (0.865)	0.062 (0.032)	0.057 (0.034)	0.065 (0.036)	0.120 (0.084)	0.876 (0.848)	0.086 (0.061)	0.100 (0.070)	0.108 (0.076)

4. D-CBG under the same sampler

Classifier time-conditioning fix (2026-07-29). The D-CBG classifiers are trained on $t \in (0,1)$ (`corrupt_mask`), but `plan_dcbg_mask` passed $t=100$ — the diffusion backbone's time scale — to the classifier at sampling time. Fixed: the classifier now receives t at its training scale, and **every D-CBG number in this record was re-measured under the fix** (the parenthesised first-hitting references from the v4 report predate it). A per-reveal-step diagnostic (`diag_dcbg_leverage.py` , 2,520 reveal steps at blk4) bounded the expected shift beforehand: the classifier's substitution signal correlates 0.96 between the two scales and the guided-vs-unguided total-variation distance at $\gamma = 1$ is 0.003 under either, so the re-measured cells move within the ± 0.012 noise floor of §2.2. The same diagnostic quantifies *why* $\gamma = 1$ guidance is inert here: the reveal posterior is near-deterministic (median top-1 probability 1.000; 18.0 nats of spread across the top-10 candidates) while the classifier's single-substitution signal spans 0.95 nats, so the tilt changes the committed token in 0.2% of reveals. $\gamma = 1$ is the exact Bayes tilt $p_\theta \cdot p_\phi$, not "guidance off" ($\gamma = 0$ is off) — the leverage, not the formula, is what is small.

4.1 Every method at $\gamma = 1$, l2r, v_Λa (seen)

blk	base	IW	adj+IW	D-CBG first-order	D-CBG exact
1	0.312 (0.300)	0.328 (0.321)	0.542 (0.540)	0.314 (0.299)	0.325 (0.321)
2	0.262 (0.238)	0.317 (0.312)	0.577 (0.401)	0.260 (0.237)	0.297 (0.264)
4	0.285 (0.262)	0.320 (0.304)	0.556 (0.328)	0.286 (0.262)	0.306 (0.281)
8	0.256 (0.214)	0.303 (0.254)	0.487 (0.273)	0.255 (0.213)	0.283 (0.219)
16	0.240 (0.194)	0.281 (0.236)	0.447 (0.250)	0.240 (0.194)	0.281 (0.207)
32	0.221 (0.160)	0.267 (0.207)	0.427 (0.215)	0.222 (0.159)	0.269 (0.174)
64	0.202 (0.119)	0.247 (0.157)	0.393 (0.162)	0.203 (0.120)	0.246 (0.126)

4.2 Every method at $\gamma = 1$, l2r, v_Λa (unseen)

blk	base	IW	adj+IW	D-CBG first-order	D-CBG exact
1	0.312 (0.300)	0.324 (0.319)	0.527 (0.530)	0.312 (0.300)	0.325 (0.319)
2	0.262 (0.238)	0.321 (0.316)	0.574 (0.415)	0.263 (0.239)	0.298 (0.266)
4	0.285 (0.262)	0.326 (0.302)	0.561 (0.331)	0.282 (0.265)	0.316 (0.280)
8	0.256 (0.214)	0.304 (0.258)	0.490 (0.273)	0.256 (0.214)	0.288 (0.222)
16	0.240 (0.194)	0.283 (0.242)	0.448 (0.254)	0.241 (0.195)	0.278 (0.207)
32	0.221 (0.160)	0.269 (0.209)	0.425 (0.219)	0.220 (0.158)	0.262 (0.166)
64	0.202 (0.119)	0.248 (0.157)	0.392 (0.162)	0.201 (0.119)	0.245 (0.125)

4.3 D-CBG exact — seen

blk	base valid	base arr	base v _Λ a	base EM	base PC	D-CBG exact valid	D-CBG exact arr	D-CBG exact v _Λ a	D-CBG exact EM	D-CBG exact PC
1	0.312 (0.302)	0.996 (0.994)	0.312 (0.300)	0.275 (0.271)	0.289 (0.283)	0.327 (0.324)	0.994 (0.995)	0.325 (0.321)	0.287 (0.283)	0.301 (0.298)
2	0.268 (0.251)	0.963 (0.949)	0.262 (0.238)	0.244 (0.233)	0.255 (0.240)	0.308 (0.278)	0.962 (0.946)	0.297 (0.264)	0.274 (0.255)	0.288 (0.264)
4	0.293 (0.279)	0.976 (0.932)	0.285 (0.262)	0.262 (0.243)	0.275 (0.258)	0.319 (0.298)	0.975 (0.939)	0.306 (0.281)	0.279 (0.258)	0.295 (0.275)
8	0.272 (0.229)	0.964 (0.938)	0.256 (0.214)	0.232 (0.202)	0.248 (0.214)	0.307 (0.239)	0.958 (0.927)	0.283 (0.219)	0.256 (0.208)	0.278 (0.222)
16	0.259 (0.206)	0.930 (0.923)	0.240 (0.194)	0.221 (0.180)	0.238 (0.190)	0.315 (0.226)	0.933 (0.917)	0.281 (0.207)	0.250 (0.197)	0.275 (0.208)
32	0.251 (0.185)	0.898 (0.870)	0.221 (0.160)	0.202 (0.157)	0.221 (0.168)	0.317 (0.206)	0.890 (0.872)	0.269 (0.174)	0.237 (0.170)	0.265 (0.184)
64	0.235 (0.141)	0.901 (0.827)	0.202 (0.119)	0.178 (0.129)	0.199 (0.134)	0.301 (0.156)	0.890 (0.813)	0.246 (0.126)	0.210 (0.132)	0.244 (0.142)

4.4 D-CBG first-order — seen

blk	base valid	base arr	base v _{LA}	base EM	base PC	D-CBG fo valid	D-CBG fo arr	D-CBG fo v _{LA}	D-CBG fo EM	D-CBG fo PC
1	0.312 (0.302)	0.996 (0.994)	0.312 (0.300)	0.275 (0.271)	0.289 (0.283)	0.314 (0.301)	0.996 (0.994)	0.314 (0.299)	0.278 (0.271)	0.291 (0.283)
2	0.268 (0.251)	0.963 (0.949)	0.262 (0.238)	0.244 (0.233)	0.255 (0.240)	0.266 (0.251)	0.963 (0.948)	0.260 (0.237)	0.243 (0.235)	0.253 (0.242)
4	0.293 (0.279)	0.976 (0.932)	0.285 (0.262)	0.262 (0.243)	0.275 (0.258)	0.294 (0.281)	0.977 (0.930)	0.286 (0.262)	0.265 (0.246)	0.277 (0.261)
8	0.272 (0.229)	0.964 (0.938)	0.256 (0.214)	0.232 (0.202)	0.248 (0.214)	0.271 (0.228)	0.964 (0.936)	0.255 (0.213)	0.231 (0.202)	0.247 (0.214)
16	0.259 (0.206)	0.930 (0.923)	0.240 (0.194)	0.221 (0.180)	0.238 (0.190)	0.259 (0.206)	0.930 (0.922)	0.240 (0.194)	0.221 (0.181)	0.238 (0.191)
32	0.251 (0.185)	0.898 (0.870)	0.221 (0.160)	0.202 (0.157)	0.221 (0.168)	0.251 (0.184)	0.899 (0.871)	0.222 (0.159)	0.202 (0.155)	0.221 (0.166)
64	0.235 (0.141)	0.901 (0.827)	0.202 (0.119)	0.178 (0.129)	0.199 (0.134)	0.236 (0.142)	0.901 (0.825)	0.203 (0.120)	0.179 (0.130)	0.200 (0.135)

- **adj+IW beats D-CBG exact by roughly 2x at a twelfth of the cost.** At blk2, 0.577 against 0.297 v_{LA}; the gap holds at every block and in both regimes. Per path that is ≈2,200 discriminator calls against ≈30,600 classifier calls (§4 cost note). Even *IW-only*, with Lemma-3 masking switched off, edges out D-CBG exact at blk2–8.
- **The first-order variant is inert under l2r too:** its v_{LA} differs from the unguided base by −0.002 to +0.002 across all seven blocks — within the ±0.012 noise floor of §2.2. The v4 report reached the same conclusion under first-hitting; the reveal order does not rescue the Taylor approximation.
- **l2r helps D-CBG, but far less, and for a different reason.** Moving to l2r lifts D-CBG exact by +0.027 to +0.120 (growing with block size) against +0.176 to +0.231 for adj+IW. D-CBG has no proposal masking — it tilts the reveal-position logits with the classifier — so the order cannot switch on a legality constraint it does not have; what it gains is only the better conditioning of a contiguous prefix, which is also what unguided base gains.

Cost accounting (from the v4 report §5.6, unchanged by the order flag, since neither the scorer-call count nor the canvas schedule depends on it): scorer calls per path are ≈2,200 for IW (L- n_{is}), ≈22 forward+backward passes for D-CBG first-order, and ≈30,600 for D-CBG exact (L-V). Wall-clock per path at blk4: 0.028 (IW) vs 0.013 (first-order) vs 0.328 (exact).

4.5 Family 0.1 — every method at $\gamma = 1$, l2r, v_{LA} (seen)

blk	base	IW	adj+IW	D-CBG first-order	D-CBG exact
1	0.111 (0.114)	0.126 (0.119)	0.229 (0.234)	0.111	0.121
2	0.089 (0.092)	0.117 (0.114)	0.281 (0.163)	0.089	0.110
4	0.103 (0.084)	0.112 (0.108)	0.227 (0.137)	0.105	0.113
8	0.086 (0.071)	0.111 (0.094)	0.199 (0.101)	0.086	0.106
16	0.081 (0.069)	0.095 (0.082)	0.187 (0.091)	0.080	0.095
32	0.066 (0.061)	0.091 (0.069)	0.180 (0.076)	0.066	0.088
64	0.062 (0.032)	0.086 (0.061)	0.173 (0.068)	0.062	0.085

5. Why block diffusion beats AR once guidance is applied

5.1 Under l2r the conditioning is AR-equivalent

Each denoising step is a fresh forward over the whole canvas, and the left-most masked position is committed; the next step's forward therefore sees that token. The conditional the model is asked for is $p(x_j \mid \text{prefix}, x_{<j})$ — the same conditional AR uses. Within-block attention is bidirectional, but the information set is identical. So l2r does not give block diffusion a better conditional than AR.

5.2 And as a sampler, AR is still the better one

Unguided v_{LA} under l2r: blk1 **0.312** > blk4 0.285 > blk2 0.262 > ... > blk64 0.202. The reversal in §3 is therefore not a sampling-quality effect; it is entirely attributable to how guidance interacts with block size.

5.3 The candidate set is a whole block, so the weights carry a lookahead

At each reveal, `_denoise_block_mask_guided` draws $n_{is} = 100$ candidates for *all* still-masked positions of the current block: `cand` has shape (b, n_{is} , block). The discriminator scores the concatenation `[committed prefix || candidate block]`, producing one weight per candidate; that

weight is then broadcast to every position of the block and accumulated into `xbar`, and only **one** position is committed. The token placed now is therefore drawn from a distribution reweighted by **how plausible the whole B-token continuation is**, not by how plausible that single token is. It is a one-ply rollout scored by the discriminator, marginalised rather than argmaxed: the tilt on the current token estimates $E[w \mid x_k = v]$, the marginal value of playing v now.

At $B = 1$ the candidate *is* the single token. There is nothing to look ahead to, and the reweighting degenerates to tilting one marginal. This is a structural property of the block, not a tuning choice, and AR cannot have it.

Two further details follow from the code. The lookahead depth is the number of positions left in the current block, so under `l2r` it is $B-1$ at the start of a block and **0 at its last position**, averaging $(B-1)/2$. And because `_disc_lengths` truncates at the destination, a candidate whose lookahead reaches the destination is scored as a *completed* path — the weight then reflects the plausibility of a finished route.

5.4 Decomposition of the guided lift (superseded in part by §5.5)

Lift in $v_{\wedge a}$ over the same block's unguided base, seen regime. "+ IW" is the discriminator reweighting with Lemma-3 masking off; "+ Lemma-3 mask" is the further gain from turning masking on.

blk	base $v_{\wedge a}$	+ IW (no mask)	+ Lemma-3 mask	adj+IW $v_{\wedge a}$
1	0.312	+0.016	+0.214	0.542
2	0.262	+0.055	+0.260	0.577
4	0.285	+0.035	+0.236	0.556
8	0.256	+0.047	+0.184	0.487
16	0.240	+0.041	+0.166	0.447
32	0.221	+0.046	+0.160	0.427
64	0.202	+0.045	+0.146	0.393

Same decomposition under first-hitting, for contrast:

blk	base $v_{\wedge a}$	+ IW (no mask)	+ Lemma-3 mask	adj+IW $v_{\wedge a}$
1	0.300	+0.021	+0.219	0.540
2	0.238	+0.074	+0.089	0.401
4	0.262	+0.042	+0.024	0.328
8	0.214	+0.040	+0.019	0.273
16	0.194	+0.042	+0.014	0.250
32	0.160	+0.047	+0.008	0.215
64	0.119	+0.038	+0.005	0.162

- **The discriminator's own lift is 3.4× larger at blk2 than at blk1** (+0.055 vs +0.016), and stays at that level for every larger block. This is the lookahead, isolated from the adjacency mask. The same pattern holds under first-hitting (+0.074 vs +0.021), so it is a property of the block, not of the order.
- **The mask's lift is what the order controls**: under first-hitting it collapses beyond blk1 (+0.024 at blk4), under `l2r` it is restored (+0.236). The reason is in the masking code: the Lemma-3 legality mask can only constrain a candidate when a neighbour of the target position is *already revealed*, and falls back to an all-ones mask otherwise. Under `l2r` the left neighbour is revealed by construction at every step; under random order many positions are committed with both neighbours still masked, hence unconstrained, and later reveals must reconcile with those free tokens. blk1 already had this property for free, which is exactly why blk1 is the one block size the order does not help.
- **Superseded in part by §5.5**. The three-arm ablation adds the missing adj-only control and shows that on $v_{\wedge a}$ the mask alone reaches nearly the adj+IW level, so the "+ Lemma-3 mask" column above should be read as the mask's contribution *given* the discriminator, not as evidence that the two are synergistic. The discriminator's own, separable contribution is on EM, and that is where the block-size signature appears — see §5.5.
- **Why the optimum is at blk2–4 rather than larger**. Candidates are drawn **mean-field**: `torch.multinomial(p_cand.reshape(b·block, V), n_is)` samples each position independently from its own marginal, and the adjacency mask constrains only the position about to be committed (the lookahead positions have masked left neighbours, so their mask degenerates). The lookahead is therefore an *unconstrained mean-field continuation scored by the discriminator*. Short blocks make that continuation plausible often enough to be informative; long blocks make it mostly nonsense, so the signal degrades — visible as ESS falling from 99.7 to 94.9 and as the mask lift decaying from +0.260 to +0.146. Making the within-block proposal ancestral is experiment E6 in §6.
- **Honest limitation**: blk1 and blk2 are separately trained checkpoints, so part of any difference is training variance. The informative structure is the *decomposition* — unguided is worse at blk2, guided is better — which points at the guidance rather than at the backbone. Seeds are experiment E3.

5.5 Three-arm ablation: adj-only / IW-only / adj+IW

The decomposition in §5.4 reads the mask's contribution as *adj+IW minus IW-only*, i.e. as an increment on top of the discriminator. This ablation adds the missing arm — **adj-only**: Lemma-3 adjacency-constrained proposals with **no discriminator at all** (`plan_guided(disc=None, adj_prop=True)`) — so each component can be read on its own. Same protocol as §3, family 0.05, seen discriminator, 1,000 reserved pairs, RAW. adj-only uses no discriminator, so it has no seen/unseen split.

first-hitting order

blk	base valid	base v _Λ a	base EM	adj only valid	adj only v _Λ a	adj only EM	IW only valid	IW only v _Λ a	IW only EM	adj+IW valid	adj+IW v _Λ a	adj+IW EM
1	0.302	0.300	0.271	0.928	0.521	0.359	0.324	0.321	0.283	0.928	0.540	0.374
2	0.251	0.238	0.233	0.441	0.365	0.346	0.328	0.312	0.306	0.492	0.401	0.388
4	0.279	0.262	0.243	0.357	0.306	0.291	0.332	0.304	0.295	0.389	0.328	0.334
8	0.229	0.214	0.202	0.278	0.242	0.231	0.294	0.254	0.270	0.331	0.273	0.295
16	0.206	0.194	0.180	0.255	0.218	0.214	0.260	0.236	0.211	0.298	0.250	0.230
32	0.185	0.160	0.157	0.228	0.181	0.184	0.244	0.207	0.204	0.265	0.215	0.219
64	0.141	0.119	0.129	0.174	0.133	0.145	0.201	0.157	0.160	0.229	0.162	0.176

left-to-right order

blk	base valid	base v _Λ a	base EM	adj only valid	adj only v _Λ a	adj only EM	IW only valid	IW only v _Λ a	IW only EM	adj+IW valid	adj+IW v _Λ a	adj+IW EM
1	0.312	0.312	0.275	0.928	0.513	0.356	0.331	0.328	0.287	0.925	0.542	0.361
2	0.268	0.262	0.244	0.866	0.547	0.395	0.322	0.317	0.292	0.867	0.577	0.440
4	0.293	0.285	0.262	0.769	0.555	0.409	0.337	0.320	0.305	0.765	0.556	0.434
8	0.272	0.256	0.232	0.761	0.476	0.366	0.321	0.303	0.289	0.760	0.487	0.392
16	0.259	0.240	0.221	0.749	0.446	0.344	0.312	0.281	0.259	0.747	0.447	0.389
32	0.251	0.221	0.202	0.720	0.427	0.336	0.322	0.267	0.259	0.723	0.427	0.376
64	0.235	0.202	0.178	0.821	0.394	0.301	0.307	0.247	0.251	0.828	0.393	0.359

family 0.1, left-to-right order

blk	base valid	base v _Λ a	base EM	adj only valid	adj only v _Λ a	adj only EM	IW only valid	IW only v _Λ a	IW only EM	adj+IW valid	adj+IW v _Λ a	adj+IW EM
1	0.112	0.111	0.103	0.921	0.218	0.135	0.127	0.126	0.110	0.918	0.229	0.147
2	0.090	0.089	0.084	0.760	0.274	0.189	0.121	0.117	0.110	0.762	0.281	0.194
4	0.109	0.103	0.095	0.609	0.226	0.157	0.116	0.112	0.103	0.610	0.227	0.174
8	0.093	0.086	0.089	0.571	0.191	0.166	0.122	0.111	0.111	0.578	0.199	0.198
16	0.091	0.081	0.078	0.636	0.185	0.170	0.113	0.095	0.097	0.658	0.187	0.202
32	0.080	0.066	0.056	0.534	0.163	0.141	0.112	0.091	0.097	0.539	0.180	0.174
64	0.076	0.062	0.057	0.748	0.163	0.139	0.120	0.086	0.100	0.774	0.173	0.173

- **The mask carries feasibility; the discriminator carries correctness.** Under l2r the discriminator adds almost nothing to v_Λa once the mask is on (−0.001 ... +0.030 across blk2–64), but it adds **+0.025 ... +0.058 EM** at every block. Adjacency masking makes paths *legal*; the discriminator makes them the *right* legal paths. Reporting only v_Λa would hide the entire contribution of the learned component.
- **The lookahead signature is in the EM increment, and it is sharp.** On top of the mask the discriminator adds +0.005 EM at blk1 (AR) against +0.025+0.058 at blk≥2 — an order of magnitude. On v_Λa the increment is +0.029 at blk1 versus −0.001+0.030 at blk≥2, i.e. flat. So what block structure buys the discriminator is not more feasible paths but better *choices among* feasible paths — exactly what a one-ply rollout scored by D should buy, and exactly what B=1 cannot have.
- **The order is what makes the mask work at B≥2.** At blk2 the mask alone lifts v_Λa by +0.285 under l2r but only +0.127 under first-hitting. blk1 is the exception in both: it is already left-to-right, so its mask arm is unchanged by the order (0.521 vs 0.513, inside the noise floor).
- **This revises the earlier reading of §5.4.** Computing the mask's contribution as *adj+IW minus IW-only* attributes the joint effect to the mask alone; the adj-only arm shows the mask reaches nearly the same v_Λa by itself. The two components are not synergistic on v_Λa — they are close to redundant there — and they are complementary on EM. The honest claim is the split above, not a synergy claim.
- **adj-only is not a hard constraint.** When the legality mask leaves no candidate with non-negligible mass the sampler falls back to the unmasked marginal (`p_cand` = where($z > 1e-9$, pm/z , `p_cand`)), present since the Lemma-3 masking was first implemented and therefore in effect for every adj+IW number in this record). That escape hatch is why adj-only validity is ~0.77–0.87 rather than 1.0: the per-edge illegality rate is only ~0.5%, but over ~24 edges it accumulates. Measuring the firing rate directly is listed as E4.

5.6 Three arms by block, family and regime

The same three arms as §5.5, laid out per block for direct quotation. All cells: left-to-right reveal order, $n_{is} = 100$, $\gamma = 1$, the 1,000 reserved except_0 OD pairs, RAW (no post-processing). Bold marks the best of the three arms within a block; the grey value in parentheses is the gain over that block's unguided base under the same sampler

(base is discriminator-independent, so it is the same number in the seen and unseen tables; its absolute values are the base columns of §3.1 and §3.5). **Adj-only uses no discriminator**, so its row is identical in the seen and unseen tables — it is repeated rather than omitted so that each table can be read on its own.

5.6.1 Edge-remove ratio 0.05 — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	Adj-only	0.928 (+0.616)	0.540 (−0.456)	0.513 (+0.201)	0.356 (+0.081)	0.406 (+0.117)
	IW-only	0.331 (+0.019)	0.992 (−0.004)	0.328 (+0.016)	0.287 (+0.012)	0.303 (+0.014)
	Adj+IW	0.925 (+0.613)	0.573 (−0.423)	0.542 (+0.230)	0.361 (+0.086)	0.415 (+0.126)
2	Adj-only	0.866 (+0.598)	0.650 (−0.313)	0.547 (+0.285)	0.395 (+0.151)	0.479 (+0.224)
	IW-only	0.322 (+0.054)	0.963 (0.000)	0.317 (+0.055)	0.292 (+0.048)	0.305 (+0.050)
	Adj+IW	0.867 (+0.599)	0.678 (−0.285)	0.577 (+0.315)	0.440 (+0.196)	0.515 (+0.260)
4	Adj-only	0.769 (+0.476)	0.734 (−0.242)	0.555 (+0.270)	0.409 (+0.147)	0.482 (+0.207)
	IW-only	0.337 (+0.044)	0.965 (−0.011)	0.320 (+0.035)	0.305 (+0.043)	0.318 (+0.043)
	Adj+IW	0.765 (+0.472)	0.732 (−0.244)	0.556 (+0.271)	0.434 (+0.172)	0.503 (+0.228)
8	Adj-only	0.761 (+0.489)	0.669 (−0.295)	0.476 (+0.220)	0.366 (+0.134)	0.456 (+0.208)
	IW-only	0.321 (+0.049)	0.963 (−0.001)	0.303 (+0.047)	0.289 (+0.057)	0.304 (+0.055)
	Adj+IW	0.760 (+0.488)	0.685 (−0.279)	0.487 (+0.231)	0.392 (+0.160)	0.480 (+0.232)
16	Adj-only	0.749 (+0.490)	0.645 (−0.285)	0.446 (+0.206)	0.344 (+0.123)	0.447 (+0.210)
	IW-only	0.312 (+0.053)	0.934 (+0.004)	0.281 (+0.041)	0.259 (+0.038)	0.280 (+0.043)
	Adj+IW	0.747 (+0.488)	0.653 (−0.277)	0.447 (+0.207)	0.389 (+0.168)	0.485 (+0.247)
32	Adj-only	0.720 (+0.469)	0.633 (−0.265)	0.427 (+0.206)	0.336 (+0.134)	0.434 (+0.214)
	IW-only	0.322 (+0.071)	0.894 (−0.004)	0.267 (+0.046)	0.259 (+0.057)	0.283 (+0.062)
	Adj+IW	0.723 (+0.472)	0.635 (−0.263)	0.427 (+0.206)	0.376 (+0.174)	0.468 (+0.247)
64	Adj-only	0.821 (+0.586)	0.524 (−0.377)	0.394 (+0.192)	0.301 (+0.123)	0.428 (+0.229)
	IW-only	0.307 (+0.072)	0.879 (−0.022)	0.247 (+0.045)	0.251 (+0.073)	0.271 (+0.072)
	Adj+IW	0.828 (+0.593)	0.511 (−0.390)	0.393 (+0.191)	0.359 (+0.181)	0.474 (+0.275)

5.6.2 Edge-remove ratio 0.05 — unseen (except 1–99 → except_0)

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	Adj-only	0.928 (+0.616)	0.540 (−0.456)	0.513 (+0.201)	0.356 (+0.081)	0.406 (+0.117)
	IW-only	0.327 (+0.015)	0.992 (−0.004)	0.324 (+0.012)	0.283 (+0.008)	0.299 (+0.010)
	Adj+IW	0.928 (+0.616)	0.556 (−0.440)	0.527 (+0.215)	0.354 (+0.079)	0.406 (+0.117)
2	Adj-only	0.866 (+0.598)	0.650 (−0.313)	0.547 (+0.285)	0.395 (+0.151)	0.479 (+0.224)
	IW-only	0.328 (+0.060)	0.966 (+0.003)	0.321 (+0.059)	0.298 (+0.054)	0.311 (+0.057)
	Adj+IW	0.863 (+0.595)	0.677 (−0.286)	0.574 (+0.312)	0.431 (+0.187)	0.507 (+0.252)
4	Adj-only	0.769 (+0.476)	0.734 (−0.242)	0.555 (+0.270)	0.409 (+0.147)	0.482 (+0.207)
	IW-only	0.336 (+0.043)	0.972 (−0.004)	0.326 (+0.041)	0.303 (+0.041)	0.316 (+0.041)
	Adj+IW	0.765 (+0.472)	0.738 (−0.238)	0.561 (+0.276)	0.439 (+0.177)	0.506 (+0.231)
8	Adj-only	0.761 (+0.489)	0.669 (−0.295)	0.476 (+0.220)	0.366 (+0.134)	0.456 (+0.208)
	IW-only	0.323 (+0.051)	0.963 (−0.001)	0.304 (+0.048)	0.290 (+0.058)	0.305 (+0.056)
	Adj+IW	0.758 (+0.486)	0.690 (−0.274)	0.490 (+0.234)	0.393 (+0.161)	0.481 (+0.232)
16	Adj-only	0.749 (+0.490)	0.645 (−0.285)	0.446 (+0.206)	0.344 (+0.123)	0.447 (+0.210)
	IW-only	0.315 (+0.056)	0.934 (+0.004)	0.283 (+0.043)	0.263 (+0.042)	0.284 (+0.046)
	Adj+IW	0.748 (+0.489)	0.653 (−0.277)	0.448 (+0.208)	0.389 (+0.168)	0.485 (+0.247)
32	Adj-only	0.720 (+0.469)	0.633 (−0.265)	0.427 (+0.206)	0.336 (+0.134)	0.434 (+0.214)
	IW-only	0.322 (+0.071)	0.896 (−0.002)	0.269 (+0.048)	0.263 (+0.061)	0.285 (+0.065)
	Adj+IW	0.723 (+0.472)	0.633 (−0.265)	0.425 (+0.204)	0.378 (+0.176)	0.469 (+0.248)
64	Adj-only	0.821 (+0.586)	0.524 (−0.377)	0.394 (+0.192)	0.301 (+0.123)	0.428 (+0.229)
	IW-only	0.306 (+0.071)	0.879 (−0.022)	0.248 (+0.046)	0.250 (+0.072)	0.270 (+0.071)
	Adj+IW	0.828 (+0.593)	0.510 (−0.391)	0.392 (+0.190)	0.359 (+0.181)	0.474 (+0.275)

5.6.3 Edge-remove ratio 0.1 — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	Adj-only	0.921 (+0.809)	0.235 (−0.758)	0.218 (+0.107)	0.135 (+0.032)	0.166 (+0.059)
	IW-only	0.127 (+0.015)	0.997 (+0.004)	0.126 (+0.015)	0.110 (+0.007)	0.116 (+0.010)
	Adj+IW	0.918 (+0.806)	0.246 (−0.747)	0.229 (+0.118)	0.147 (+0.044)	0.180 (+0.073)
2	Adj-only	0.760 (+0.670)	0.377 (−0.597)	0.274 (+0.185)	0.189 (+0.105)	0.242 (+0.156)
	IW-only	0.121 (+0.031)	0.967 (−0.007)	0.117 (+0.028)	0.110 (+0.026)	0.114 (+0.028)
	Adj+IW	0.762 (+0.672)	0.382 (−0.592)	0.281 (+0.192)	0.194 (+0.110)	0.257 (+0.171)
4	Adj-only	0.609 (+0.500)	0.414 (−0.560)	0.226 (+0.123)	0.157 (+0.062)	0.203 (+0.103)
	IW-only	0.116 (+0.007)	0.979 (+0.005)	0.112 (+0.009)	0.103 (+0.008)	0.108 (+0.008)
	Adj+IW	0.610 (+0.501)	0.410 (−0.564)	0.227 (+0.124)	0.174 (+0.079)	0.219 (+0.119)
8	Adj-only	0.571 (+0.478)	0.433 (−0.532)	0.191 (+0.105)	0.166 (+0.077)	0.228 (+0.138)
	IW-only	0.122 (+0.029)	0.961 (−0.004)	0.111 (+0.025)	0.111 (+0.022)	0.116 (+0.025)
	Adj+IW	0.578 (+0.485)	0.422 (−0.543)	0.199 (+0.113)	0.198 (+0.109)	0.263 (+0.172)
16	Adj-only	0.636 (+0.545)	0.382 (−0.549)	0.185 (+0.104)	0.170 (+0.092)	0.259 (+0.175)
	IW-only	0.113 (+0.022)	0.928 (−0.003)	0.095 (+0.014)	0.097 (+0.019)	0.105 (+0.021)
	Adj+IW	0.658 (+0.567)	0.364 (−0.567)	0.187 (+0.106)	0.202 (+0.124)	0.298 (+0.214)
32	Adj-only	0.534 (+0.454)	0.407 (−0.479)	0.163 (+0.097)	0.141 (+0.085)	0.213 (+0.148)
	IW-only	0.112 (+0.032)	0.897 (+0.011)	0.091 (+0.025)	0.097 (+0.041)	0.103 (+0.037)
	Adj+IW	0.539 (+0.459)	0.411 (−0.475)	0.180 (+0.114)	0.174 (+0.118)	0.248 (+0.182)
64	Adj-only	0.748 (+0.672)	0.292 (−0.608)	0.163 (+0.101)	0.139 (+0.082)	0.242 (+0.177)
	IW-only	0.120 (+0.044)	0.876 (−0.024)	0.086 (+0.024)	0.100 (+0.043)	0.108 (+0.042)
	Adj+IW	0.774 (+0.698)	0.282 (−0.618)	0.173 (+0.111)	0.173 (+0.116)	0.287 (+0.222)

5.6.4 Edge-remove ratio 0.1 — unseen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	Adj-only	0.921 (+0.809)	0.235 (−0.758)	0.218 (+0.107)	0.135 (+0.032)	0.166 (+0.059)
	IW-only	0.129 (+0.017)	0.995 (+0.002)	0.128 (+0.017)	0.117 (+0.014)	0.121 (+0.015)
	Adj+IW	0.920 (+0.808)	0.244 (−0.749)	0.228 (+0.117)	0.148 (+0.045)	0.181 (+0.074)
2	Adj-only	0.760 (+0.670)	0.377 (−0.597)	0.274 (+0.185)	0.189 (+0.105)	0.242 (+0.156)
	IW-only	0.121 (+0.031)	0.965 (−0.009)	0.116 (+0.027)	0.109 (+0.025)	0.114 (+0.027)
	Adj+IW	0.764 (+0.674)	0.377 (−0.597)	0.280 (+0.191)	0.190 (+0.106)	0.253 (+0.167)
4	Adj-only	0.609 (+0.500)	0.414 (−0.560)	0.226 (+0.123)	0.157 (+0.062)	0.203 (+0.103)
	IW-only	0.123 (+0.014)	0.965 (−0.009)	0.116 (+0.013)	0.109 (+0.014)	0.114 (+0.014)
	Adj+IW	0.618 (+0.509)	0.405 (−0.569)	0.226 (+0.123)	0.169 (+0.074)	0.216 (+0.116)
8	Adj-only	0.571 (+0.478)	0.433 (−0.532)	0.191 (+0.105)	0.166 (+0.077)	0.228 (+0.138)
	IW-only	0.122 (+0.029)	0.963 (−0.002)	0.112 (+0.026)	0.114 (+0.025)	0.117 (+0.027)
	Adj+IW	0.579 (+0.486)	0.422 (−0.543)	0.198 (+0.112)	0.193 (+0.104)	0.262 (+0.172)
16	Adj-only	0.636 (+0.545)	0.382 (−0.549)	0.185 (+0.104)	0.170 (+0.092)	0.259 (+0.175)
	IW-only	0.117 (+0.026)	0.928 (−0.003)	0.098 (+0.017)	0.099 (+0.021)	0.107 (+0.023)
	Adj+IW	0.653 (+0.562)	0.361 (−0.570)	0.181 (+0.100)	0.198 (+0.120)	0.297 (+0.212)
32	Adj-only	0.534 (+0.454)	0.407 (−0.479)	0.163 (+0.097)	0.141 (+0.085)	0.213 (+0.148)
	IW-only	0.114 (+0.034)	0.898 (+0.012)	0.093 (+0.027)	0.098 (+0.042)	0.104 (+0.038)
	Adj+IW	0.540 (+0.460)	0.408 (−0.478)	0.182 (+0.116)	0.170 (+0.114)	0.245 (+0.180)
64	Adj-only	0.748 (+0.672)	0.292 (−0.608)	0.163 (+0.101)	0.139 (+0.082)	0.242 (+0.177)
	IW-only	0.120 (+0.044)	0.878 (−0.022)	0.086 (+0.024)	0.100 (+0.043)	0.107 (+0.042)
	Adj+IW	0.774 (+0.698)	0.282 (−0.618)	0.174 (+0.112)	0.171 (+0.114)	0.286 (+0.221)

5.7 D-CBG (first-order) / D-CBG (exact) / Adj+IW, by block

The published baseline in both of its variants against our method, in the §5.6 layout. Same conditions throughout: left-to-right reveal order, $\gamma = 1$ for both methods, the 1,000 reserved except_0 OD pairs, RAW. Parentheses are the gain over that block's unguided base; bold is the best of the three within a block. Cost per path is not equal across rows and is what makes the comparison interesting: ≈ 22 classifier forward+backward passes for first-order, $\approx 30,600$ classifier calls for exact (L-V), $\approx 2,200$ discriminator calls for Adj+IW ($L \cdot n_{\text{IS}}$).

5.7.1 Edge-remove ratio 0.05 — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.314 (+0.002)	0.996 (0.000)	0.314 (+0.002)	0.278 (+0.003)	0.291 (+0.002)
	D-CBG (exact)	0.327 (+0.015)	0.994 (−0.002)	0.325 (+0.013)	0.287 (+0.012)	0.301 (+0.012)
	Adj+IW	0.925 (+0.613)	0.573 (−0.423)	0.542 (+0.230)	0.361 (+0.086)	0.415 (+0.126)
2	D-CBG (first-order)	0.266 (−0.002)	0.963 (0.000)	0.260 (−0.002)	0.243 (−0.001)	0.253 (−0.002)
	D-CBG (exact)	0.308 (+0.040)	0.962 (−0.001)	0.297 (+0.035)	0.274 (+0.030)	0.288 (+0.033)
	Adj+IW	0.867 (+0.599)	0.678 (−0.285)	0.577 (+0.315)	0.440 (+0.196)	0.515 (+0.260)
4	D-CBG (first-order)	0.294 (+0.001)	0.977 (+0.001)	0.286 (+0.001)	0.265 (+0.003)	0.277 (+0.002)
	D-CBG (exact)	0.319 (+0.026)	0.975 (−0.001)	0.306 (+0.021)	0.279 (+0.017)	0.295 (+0.020)
	Adj+IW	0.765 (+0.472)	0.732 (−0.244)	0.556 (+0.271)	0.434 (+0.172)	0.503 (+0.228)
8	D-CBG (first-order)	0.271 (−0.001)	0.964 (0.000)	0.255 (−0.001)	0.231 (−0.001)	0.247 (−0.001)
	D-CBG (exact)	0.307 (+0.035)	0.958 (−0.006)	0.283 (+0.027)	0.256 (+0.024)	0.278 (+0.029)
	Adj+IW	0.760 (+0.488)	0.685 (−0.279)	0.487 (+0.231)	0.392 (+0.160)	0.480 (+0.232)
16	D-CBG (first-order)	0.259 (0.000)	0.930 (0.000)	0.240 (0.000)	0.221 (0.000)	0.238 (0.000)
	D-CBG (exact)	0.315 (+0.056)	0.933 (+0.003)	0.281 (+0.041)	0.250 (+0.029)	0.275 (+0.038)
	Adj+IW	0.747 (+0.488)	0.653 (−0.277)	0.447 (+0.207)	0.389 (+0.168)	0.485 (+0.247)
32	D-CBG (first-order)	0.251 (0.000)	0.899 (+0.001)	0.222 (+0.001)	0.202 (0.000)	0.221 (0.000)
	D-CBG (exact)	0.317 (+0.066)	0.890 (−0.008)	0.269 (+0.048)	0.237 (+0.035)	0.265 (+0.044)
	Adj+IW	0.723 (+0.472)	0.635 (−0.263)	0.427 (+0.206)	0.376 (+0.174)	0.468 (+0.247)
64	D-CBG (first-order)	0.236 (+0.001)	0.901 (0.000)	0.203 (+0.001)	0.179 (+0.001)	0.200 (+0.001)
	D-CBG (exact)	0.301 (+0.066)	0.890 (−0.011)	0.246 (+0.044)	0.210 (+0.032)	0.244 (+0.045)
	Adj+IW	0.828 (+0.593)	0.511 (−0.390)	0.393 (+0.191)	0.359 (+0.181)	0.474 (+0.275)

5.7.2 Edge-remove ratio 0.05 — unseen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.312 (0.000)	0.997 (+0.001)	0.312 (0.000)	0.275 (0.000)	0.289 (0.000)
	D-CBG (exact)	0.327 (+0.015)	0.994 (−0.002)	0.325 (+0.013)	0.284 (+0.009)	0.300 (+0.011)
	Adj+IW	0.928 (+0.616)	0.556 (−0.440)	0.527 (+0.215)	0.354 (+0.079)	0.406 (+0.117)
2	D-CBG (first-order)	0.269 (+0.001)	0.963 (0.000)	0.263 (+0.001)	0.245 (+0.001)	0.256 (+0.001)
	D-CBG (exact)	0.310 (+0.042)	0.962 (−0.001)	0.298 (+0.036)	0.275 (+0.031)	0.289 (+0.035)
	Adj+IW	0.863 (+0.595)	0.677 (−0.286)	0.574 (+0.312)	0.431 (+0.187)	0.507 (+0.252)
4	D-CBG (first-order)	0.290 (−0.003)	0.976 (0.000)	0.282 (−0.003)	0.263 (+0.001)	0.275 (0.000)
	D-CBG (exact)	0.328 (+0.035)	0.974 (−0.002)	0.316 (+0.031)	0.283 (+0.021)	0.301 (+0.026)
	Adj+IW	0.765 (+0.472)	0.738 (−0.238)	0.561 (+0.276)	0.439 (+0.177)	0.506 (+0.231)
8	D-CBG (first-order)	0.272 (0.000)	0.964 (0.000)	0.256 (0.000)	0.232 (0.000)	0.248 (0.000)
	D-CBG (exact)	0.314 (+0.042)	0.959 (−0.005)	0.288 (+0.032)	0.254 (+0.022)	0.278 (+0.030)
	Adj+IW	0.758 (+0.486)	0.690 (−0.274)	0.490 (+0.234)	0.393 (+0.161)	0.481 (+0.232)
16	D-CBG (first-order)	0.260 (+0.001)	0.931 (+0.001)	0.241 (+0.001)	0.222 (+0.001)	0.239 (+0.001)
	D-CBG (exact)	0.312 (+0.053)	0.933 (+0.003)	0.278 (+0.038)	0.249 (+0.028)	0.273 (+0.036)
	Adj+IW	0.748 (+0.489)	0.653 (−0.277)	0.448 (+0.208)	0.389 (+0.168)	0.485 (+0.247)
32	D-CBG (first-order)	0.250 (−0.001)	0.897 (−0.001)	0.220 (−0.001)	0.201 (−0.001)	0.220 (−0.001)
	D-CBG (exact)	0.314 (+0.063)	0.885 (−0.013)	0.262 (+0.041)	0.233 (+0.031)	0.262 (+0.042)
	Adj+IW	0.723 (+0.472)	0.633 (−0.265)	0.425 (+0.204)	0.378 (+0.176)	0.469 (+0.248)
64	D-CBG (first-order)	0.234 (−0.001)	0.901 (0.000)	0.201 (−0.001)	0.177 (−0.001)	0.198 (−0.001)
	D-CBG (exact)	0.299 (+0.064)	0.896 (−0.005)	0.245 (+0.043)	0.210 (+0.032)	0.243 (+0.044)
	Adj+IW	0.828 (+0.593)	0.510 (−0.391)	0.392 (+0.190)	0.359 (+0.181)	0.474 (+0.275)

5.7.3 Edge-remove ratio 0.1 — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.112 (0.000)	0.993 (0.000)	0.111 (0.000)	0.102 (−0.001)	0.106 (−0.001)
	D-CBG (exact)	0.123 (+0.011)	0.993 (0.000)	0.121 (+0.010)	0.109 (+0.006)	0.113 (+0.007)
	Adj+IW	0.918 (+0.806)	0.246 (−0.747)	0.229 (+0.118)	0.147 (+0.044)	0.180 (+0.073)
2	D-CBG (first-order)	0.090 (0.000)	0.974 (0.000)	0.089 (0.000)	0.084 (0.000)	0.086 (0.000)
	D-CBG (exact)	0.113 (+0.023)	0.976 (+0.002)	0.110 (+0.021)	0.101 (+0.017)	0.106 (+0.019)
	Adj+IW	0.762 (+0.672)	0.382 (−0.592)	0.281 (+0.192)	0.194 (+0.110)	0.257 (+0.171)
4	D-CBG (first-order)	0.111 (+0.002)	0.974 (0.000)	0.105 (+0.002)	0.097 (+0.002)	0.102 (+0.002)
	D-CBG (exact)	0.118 (+0.009)	0.972 (−0.002)	0.113 (+0.010)	0.107 (+0.012)	0.111 (+0.010)
	Adj+IW	0.610 (+0.501)	0.410 (−0.564)	0.227 (+0.124)	0.174 (+0.079)	0.219 (+0.119)
8	D-CBG (first-order)	0.093 (0.000)	0.965 (0.000)	0.086 (0.000)	0.089 (0.000)	0.090 (0.000)
	D-CBG (exact)	0.119 (+0.026)	0.958 (−0.007)	0.106 (+0.020)	0.109 (+0.020)	0.114 (+0.023)
	Adj+IW	0.578 (+0.485)	0.422 (−0.543)	0.199 (+0.113)	0.198 (+0.109)	0.263 (+0.172)
16	D-CBG (first-order)	0.091 (0.000)	0.931 (0.000)	0.080 (−0.001)	0.077 (−0.001)	0.084 (−0.001)
	D-CBG (exact)	0.115 (+0.024)	0.934 (+0.003)	0.095 (+0.014)	0.092 (+0.014)	0.102 (+0.018)
	Adj+IW	0.658 (+0.567)	0.364 (−0.567)	0.187 (+0.106)	0.202 (+0.124)	0.298 (+0.214)
32	D-CBG (first-order)	0.080 (0.000)	0.885 (−0.001)	0.066 (0.000)	0.056 (0.000)	0.066 (0.000)
	D-CBG (exact)	0.114 (+0.034)	0.882 (−0.004)	0.088 (+0.022)	0.078 (+0.022)	0.091 (+0.026)
	Adj+IW	0.539 (+0.459)	0.411 (−0.475)	0.180 (+0.114)	0.174 (+0.118)	0.248 (+0.182)
64	D-CBG (first-order)	0.076 (0.000)	0.900 (0.000)	0.062 (0.000)	0.057 (0.000)	0.065 (0.000)
	D-CBG (exact)	0.119 (+0.043)	0.892 (−0.008)	0.085 (+0.023)	0.082 (+0.025)	0.095 (+0.030)
	Adj+IW	0.774 (+0.698)	0.282 (−0.618)	0.173 (+0.111)	0.173 (+0.116)	0.287 (+0.222)

5.7.4 Edge-remove ratio 0.1 — unseen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.112 (0.000)	0.993 (0.000)	0.111 (0.000)	0.103 (0.000)	0.106 (0.000)
	D-CBG (exact)	0.120 (+0.008)	0.992 (−0.001)	0.117 (+0.006)	0.108 (+0.005)	0.112 (+0.005)
	Adj+IW	0.920 (+0.808)	0.244 (−0.749)	0.228 (+0.117)	0.148 (+0.045)	0.181 (+0.074)
2	D-CBG (first-order)	0.090 (0.000)	0.974 (0.000)	0.089 (0.000)	0.084 (0.000)	0.086 (0.000)
	D-CBG (exact)	0.115 (+0.025)	0.974 (0.000)	0.111 (+0.022)	0.101 (+0.017)	0.107 (+0.020)
	Adj+IW	0.764 (+0.674)	0.377 (−0.597)	0.280 (+0.191)	0.190 (+0.106)	0.253 (+0.167)
4	D-CBG (first-order)	0.110 (+0.001)	0.974 (0.000)	0.104 (+0.001)	0.095 (0.000)	0.100 (0.000)
	D-CBG (exact)	0.118 (+0.009)	0.973 (−0.001)	0.113 (+0.010)	0.106 (+0.011)	0.110 (+0.010)
	Adj+IW	0.618 (+0.509)	0.405 (−0.569)	0.226 (+0.123)	0.169 (+0.074)	0.216 (+0.116)
8	D-CBG (first-order)	0.091 (−0.002)	0.965 (0.000)	0.085 (−0.001)	0.086 (−0.003)	0.088 (−0.003)
	D-CBG (exact)	0.116 (+0.023)	0.956 (−0.009)	0.103 (+0.017)	0.107 (+0.018)	0.111 (+0.021)
	Adj+IW	0.579 (+0.486)	0.422 (−0.543)	0.198 (+0.112)	0.193 (+0.104)	0.262 (+0.172)
16	D-CBG (first-order)	0.091 (0.000)	0.930 (−0.001)	0.081 (0.000)	0.078 (0.000)	0.084 (0.000)
	D-CBG (exact)	0.117 (+0.026)	0.936 (+0.005)	0.097 (+0.016)	0.093 (+0.015)	0.103 (+0.019)
	Adj+IW	0.653 (+0.562)	0.361 (−0.570)	0.181 (+0.100)	0.198 (+0.120)	0.297 (+0.212)
32	D-CBG (first-order)	0.080 (0.000)	0.886 (0.000)	0.066 (0.000)	0.056 (0.000)	0.066 (0.000)
	D-CBG (exact)	0.118 (+0.038)	0.877 (−0.009)	0.089 (+0.023)	0.079 (+0.023)	0.093 (+0.027)
	Adj+IW	0.540 (+0.460)	0.408 (−0.478)	0.182 (+0.116)	0.170 (+0.114)	0.245 (+0.180)
64	D-CBG (first-order)	0.076 (0.000)	0.899 (−0.001)	0.061 (−0.001)	0.057 (0.000)	0.065 (0.000)
	D-CBG (exact)	0.123 (+0.047)	0.889 (−0.011)	0.087 (+0.025)	0.085 (+0.028)	0.099 (+0.034)
	Adj+IW	0.774 (+0.698)	0.282 (−0.618)	0.174 (+0.112)	0.171 (+0.114)	0.286 (+0.221)

5.8 D-CBG at $\gamma = 4$, by block

The §5.7 layout repeated with $\gamma = 4$ for the **D-CBG rows** (the γ giving the strongest exact-variant response in the leverage diagnostic of §4 without collapsing arrival). Everything else is unchanged: left-to-right reveal order, family 0.05, the 1,000 reserved except_0 OD pairs, RAW, classifier time-conditioning fix applied. The Adj+IW reference row is copied from §5.7 and still uses $w_\gamma = 1$ — the γ sweep was granted to the baseline only. Parentheses: gain over that block's unguided base; bold: best of the three arms in the block.

5.8.1 Edge-remove ratio 0.05 — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.313 (+0.001)	0.996 (0.000)	0.313 (+0.001)	0.278 (+0.003)	0.291 (+0.002)
	D-CBG (exact)	0.388 (+0.076)	0.981 (−0.015)	0.384 (+0.072)	0.317 (+0.042)	0.341 (+0.052)
	Adj+IW	0.925 (+0.613)	0.573 (−0.423)	0.542 (+0.230)	0.361 (+0.086)	0.415 (+0.126)
2	D-CBG (first-order)	0.268 (0.000)	0.965 (+0.002)	0.264 (+0.002)	0.247 (+0.003)	0.256 (+0.002)
	D-CBG (exact)	0.398 (+0.130)	0.951 (−0.012)	0.372 (+0.110)	0.326 (+0.082)	0.352 (+0.097)
	Adj+IW	0.867 (+0.599)	0.678 (−0.285)	0.577 (+0.315)	0.440 (+0.196)	0.515 (+0.260)
4	D-CBG (first-order)	0.296 (+0.003)	0.975 (−0.001)	0.287 (+0.002)	0.264 (+0.002)	0.277 (+0.002)
	D-CBG (exact)	0.398 (+0.105)	0.950 (−0.026)	0.363 (+0.078)	0.322 (+0.060)	0.351 (+0.076)
	Adj+IW	0.765 (+0.472)	0.732 (−0.244)	0.556 (+0.271)	0.434 (+0.172)	0.503 (+0.228)
8	D-CBG (first-order)	0.275 (+0.003)	0.961 (−0.003)	0.257 (+0.001)	0.234 (+0.002)	0.251 (+0.003)
	D-CBG (exact)	0.398 (+0.126)	0.917 (−0.047)	0.341 (+0.085)	0.305 (+0.073)	0.340 (+0.092)
	Adj+IW	0.760 (+0.488)	0.685 (−0.279)	0.487 (+0.231)	0.392 (+0.160)	0.480 (+0.232)
16	D-CBG (first-order)	0.260 (+0.001)	0.928 (−0.002)	0.240 (0.000)	0.222 (+0.001)	0.239 (+0.001)
	D-CBG (exact)	0.462 (+0.203)	0.876 (−0.054)	0.355 (+0.115)	0.315 (+0.094)	0.365 (+0.128)
	Adj+IW	0.747 (+0.488)	0.653 (−0.277)	0.447 (+0.207)	0.389 (+0.168)	0.485 (+0.247)
32	D-CBG (first-order)	0.254 (+0.003)	0.897 (−0.001)	0.223 (+0.002)	0.203 (+0.001)	0.222 (+0.002)
	D-CBG (exact)	0.485 (+0.234)	0.826 (−0.072)	0.356 (+0.135)	0.306 (+0.104)	0.364 (+0.143)
	Adj+IW	0.723 (+0.472)	0.635 (−0.263)	0.427 (+0.206)	0.376 (+0.174)	0.468 (+0.247)
64	D-CBG (first-order)	0.235 (0.000)	0.902 (+0.001)	0.202 (0.000)	0.179 (+0.001)	0.200 (+0.001)
	D-CBG (exact)	0.506 (+0.271)	0.790 (−0.111)	0.345 (+0.143)	0.290 (+0.112)	0.360 (+0.161)
	Adj+IW	0.828 (+0.593)	0.511 (−0.390)	0.393 (+0.191)	0.359 (+0.181)	0.474 (+0.275)

5.8.2 Edge-remove ratio 0.05 — unseen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.314 (+0.002)	0.997 (+0.001)	0.314 (+0.002)	0.275 (0.000)	0.290 (+0.001)
	D-CBG (exact)	0.393 (+0.081)	0.980 (−0.016)	0.387 (+0.075)	0.317 (+0.042)	0.340 (+0.051)
	Adj+IW	0.928 (+0.616)	0.556 (−0.440)	0.527 (+0.215)	0.354 (+0.079)	0.406 (+0.117)
2	D-CBG (first-order)	0.268 (0.000)	0.963 (0.000)	0.262 (0.000)	0.245 (+0.001)	0.255 (+0.001)
	D-CBG (exact)	0.397 (+0.129)	0.947 (−0.016)	0.365 (+0.103)	0.315 (+0.071)	0.343 (+0.088)
	Adj+IW	0.863 (+0.595)	0.677 (−0.286)	0.574 (+0.312)	0.431 (+0.187)	0.507 (+0.252)
4	D-CBG (first-order)	0.290 (−0.003)	0.976 (0.000)	0.282 (−0.003)	0.263 (+0.001)	0.275 (0.000)
	D-CBG (exact)	0.403 (+0.110)	0.945 (−0.031)	0.366 (+0.081)	0.316 (+0.054)	0.348 (+0.073)
	Adj+IW	0.765 (+0.472)	0.738 (−0.238)	0.561 (+0.276)	0.439 (+0.177)	0.506 (+0.231)
8	D-CBG (first-order)	0.270 (−0.002)	0.963 (−0.001)	0.254 (−0.002)	0.230 (−0.002)	0.246 (−0.002)
	D-CBG (exact)	0.441 (+0.169)	0.910 (−0.054)	0.367 (+0.111)	0.316 (+0.084)	0.359 (+0.111)
	Adj+IW	0.758 (+0.486)	0.690 (−0.274)	0.490 (+0.234)	0.393 (+0.161)	0.481 (+0.232)
16	D-CBG (first-order)	0.262 (+0.003)	0.932 (+0.002)	0.242 (+0.002)	0.224 (+0.003)	0.241 (+0.003)
	D-CBG (exact)	0.443 (+0.184)	0.867 (−0.063)	0.336 (+0.096)	0.297 (+0.076)	0.345 (+0.108)
	Adj+IW	0.748 (+0.489)	0.653 (−0.277)	0.448 (+0.208)	0.389 (+0.168)	0.485 (+0.247)
32	D-CBG (first-order)	0.251 (0.000)	0.896 (−0.002)	0.221 (0.000)	0.202 (0.000)	0.221 (0.000)
	D-CBG (exact)	0.466 (+0.215)	0.823 (−0.075)	0.327 (+0.106)	0.297 (+0.095)	0.352 (+0.131)
	Adj+IW	0.723 (+0.472)	0.633 (−0.265)	0.425 (+0.204)	0.378 (+0.176)	0.469 (+0.248)
64	D-CBG (first-order)	0.235 (0.000)	0.902 (+0.001)	0.202 (0.000)	0.178 (0.000)	0.200 (0.000)
	D-CBG (exact)	0.526 (+0.291)	0.768 (−0.133)	0.336 (+0.134)	0.279 (+0.101)	0.356 (+0.157)
	Adj+IW	0.828 (+0.593)	0.510 (−0.391)	0.392 (+0.190)	0.359 (+0.181)	0.474 (+0.275)

5.8.3 Edge-remove ratio 0.1 — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.109 (−0.003)	0.993 (0.000)	0.108 (−0.003)	0.101 (−0.002)	0.104 (−0.003)
	D-CBG (exact)	0.150 (+0.038)	0.957 (−0.036)	0.147 (+0.036)	0.124 (+0.021)	0.132 (+0.026)
	Adj+IW	0.918 (+0.806)	0.246 (−0.747)	0.229 (+0.118)	0.147 (+0.044)	0.180 (+0.073)
2	D-CBG (first-order)	0.090 (0.000)	0.973 (−0.001)	0.089 (0.000)	0.084 (0.000)	0.086 (0.000)
	D-CBG (exact)	0.197 (+0.107)	0.931 (−0.043)	0.168 (+0.079)	0.135 (+0.051)	0.152 (+0.066)
	Adj+IW	0.762 (+0.672)	0.382 (−0.592)	0.281 (+0.192)	0.194 (+0.110)	0.257 (+0.171)
4	D-CBG (first-order)	0.108 (−0.001)	0.974 (0.000)	0.102 (−0.001)	0.095 (0.000)	0.100 (−0.001)
	D-CBG (exact)	0.155 (+0.046)	0.954 (−0.020)	0.142 (+0.039)	0.132 (+0.037)	0.139 (+0.039)
	Adj+IW	0.610 (+0.501)	0.410 (−0.564)	0.227 (+0.124)	0.174 (+0.079)	0.219 (+0.119)
8	D-CBG (first-order)	0.097 (+0.004)	0.964 (−0.001)	0.089 (+0.003)	0.092 (+0.003)	0.094 (+0.003)
	D-CBG (exact)	0.207 (+0.114)	0.910 (−0.055)	0.162 (+0.076)	0.151 (+0.062)	0.169 (+0.079)
	Adj+IW	0.578 (+0.485)	0.422 (−0.543)	0.199 (+0.113)	0.198 (+0.109)	0.263 (+0.172)
16	D-CBG (first-order)	0.090 (−0.001)	0.931 (0.000)	0.079 (−0.002)	0.076 (−0.002)	0.083 (−0.002)
	D-CBG (exact)	0.230 (+0.139)	0.857 (−0.074)	0.145 (+0.064)	0.135 (+0.057)	0.165 (+0.081)
	Adj+IW	0.658 (+0.567)	0.364 (−0.567)	0.187 (+0.106)	0.202 (+0.124)	0.298 (+0.214)
32	D-CBG (first-order)	0.080 (0.000)	0.885 (−0.001)	0.065 (−0.001)	0.057 (+0.001)	0.066 (+0.001)
	D-CBG (exact)	0.243 (+0.163)	0.807 (−0.079)	0.133 (+0.067)	0.128 (+0.072)	0.162 (+0.096)
	Adj+IW	0.539 (+0.459)	0.411 (−0.475)	0.180 (+0.114)	0.174 (+0.118)	0.248 (+0.182)
64	D-CBG (first-order)	0.078 (+0.002)	0.900 (0.000)	0.063 (+0.001)	0.058 (+0.001)	0.067 (+0.002)
	D-CBG (exact)	0.270 (+0.194)	0.760 (−0.140)	0.132 (+0.070)	0.135 (+0.078)	0.173 (+0.108)
	Adj+IW	0.774 (+0.698)	0.282 (−0.618)	0.173 (+0.111)	0.173 (+0.116)	0.287 (+0.222)

5.8.4 Edge-remove ratio 0.1 — unseen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.111 (−0.001)	0.993 (0.000)	0.110 (−0.001)	0.101 (−0.002)	0.105 (−0.002)
	D-CBG (exact)	0.162 (+0.050)	0.924 (−0.069)	0.156 (+0.045)	0.124 (+0.021)	0.135 (+0.029)
	Adj+IW	0.920 (+0.808)	0.244 (−0.749)	0.228 (+0.117)	0.148 (+0.045)	0.181 (+0.074)
2	D-CBG (first-order)	0.091 (+0.001)	0.974 (0.000)	0.090 (+0.001)	0.084 (0.000)	0.087 (+0.001)
	D-CBG (exact)	0.199 (+0.109)	0.918 (−0.056)	0.159 (+0.070)	0.139 (+0.055)	0.156 (+0.070)
	Adj+IW	0.764 (+0.674)	0.377 (−0.597)	0.280 (+0.191)	0.190 (+0.106)	0.253 (+0.167)
4	D-CBG (first-order)	0.110 (+0.001)	0.973 (−0.001)	0.103 (0.000)	0.095 (0.000)	0.101 (+0.001)
	D-CBG (exact)	0.190 (+0.081)	0.921 (−0.053)	0.147 (+0.044)	0.134 (+0.039)	0.151 (+0.051)
	Adj+IW	0.618 (+0.509)	0.405 (−0.569)	0.226 (+0.123)	0.169 (+0.074)	0.216 (+0.116)
8	D-CBG (first-order)	0.093 (0.000)	0.964 (−0.001)	0.087 (+0.001)	0.088 (−0.001)	0.090 (−0.001)
	D-CBG (exact)	0.226 (+0.133)	0.885 (−0.080)	0.149 (+0.063)	0.144 (+0.055)	0.169 (+0.079)
	Adj+IW	0.579 (+0.486)	0.422 (−0.543)	0.198 (+0.112)	0.193 (+0.104)	0.262 (+0.172)
16	D-CBG (first-order)	0.090 (−0.001)	0.930 (−0.001)	0.079 (−0.002)	0.077 (−0.001)	0.083 (−0.001)
	D-CBG (exact)	0.260 (+0.169)	0.830 (−0.101)	0.141 (+0.060)	0.141 (+0.063)	0.177 (+0.093)
	Adj+IW	0.653 (+0.562)	0.361 (−0.570)	0.181 (+0.100)	0.198 (+0.120)	0.297 (+0.212)
32	D-CBG (first-order)	0.080 (0.000)	0.885 (−0.001)	0.066 (0.000)	0.056 (0.000)	0.066 (0.000)
	D-CBG (exact)	0.269 (+0.189)	0.786 (−0.100)	0.130 (+0.064)	0.123 (+0.067)	0.159 (+0.094)
	Adj+IW	0.540 (+0.460)	0.408 (−0.478)	0.182 (+0.116)	0.170 (+0.114)	0.245 (+0.180)
64	D-CBG (first-order)	0.077 (+0.001)	0.898 (−0.002)	0.061 (−0.001)	0.058 (+0.001)	0.066 (+0.001)
	D-CBG (exact)	0.338 (+0.262)	0.724 (−0.176)	0.134 (+0.072)	0.133 (+0.076)	0.181 (+0.116)
	Adj+IW	0.774 (+0.698)	0.282 (−0.618)	0.174 (+0.112)	0.171 (+0.114)	0.286 (+0.221)

- **Exact D-CBG responds to γ ; the response is the predicted validity/arrival trade.** Over $\gamma = 1$ it gains +0.057 to +0.099 $\vee \Delta$ across the seven blocks (largest at blk64: validity 0.301 $\rightarrow$ 0.506, arrival 0.890 $\rightarrow$ 0.790, product up +0.099). The per-step diagnostic of §4 predicted this margin of action: $\gamma = 4$ flips the committed token at $\approx 1\%$ of reveals, ≈ 22 reveals per path make that $\approx 20\%$ of paths touched, and the flips concentrate on the near-tie steps.
- **The first-order variant stays inert at $\gamma = 4$** (−0.001 to +0.004 $\vee \Delta$) — the Taylor signal, unlike the enumeration, has no leverage for γ to amplify. Raising γ rescues the mechanism only where a mechanism exists.
- **The ranking does not change, but the reading sharpens.** Adj+IW (still at $\gamma = 1$) leads exact D-CBG ($\gamma = 4$) at every block — by 0.205 $\vee \Delta$ at blk2, narrowing to 0.048 at blk64 — while costing a twelfth as much per path. Note the asymmetry of the comparison: γ was tuned for D-CBG only; the IW side's w_γ remains 1.0 throughout this record.
- **Family 0.1 replicates the pattern.** Exact D-CBG gains +0.026 to +0.058 $\vee \Delta$ over $\gamma = 1$ (seen), and Adj+IW at $w_\gamma = 1$ stays ahead at every block (0.037–0.113 $\vee \Delta$).

5.9 Adj+D-CBG: the candidate restriction ported to the baseline

The Lemma-3 adjacency restriction of §5.5 applied to the D-CBG sampler: at each reveal the candidate set is restricted to the scenario-legal successors of the (revealed, under l2r) left neighbour, END stays available, and a row whose legal set carries no model mass falls back to the unmasked distribution — the same rule, ported verbatim (`dcbg_plugin.py::plan_dcbg_mask` , `adj_scn`). The classifier then tilts only the surviving candidates. For the exact variant this also collapses the cost: enumeration scores legal candidates only, $\approx L \cdot (\deg+2) \approx 10^2$ classifier calls per path instead of $\approx 30,600$ — cheaper than Adj+IW's $\approx 2,200$. Layout of §5.7; Adj+IW reference row unchanged ($w_\gamma = 1$); family 0.05.

5.9.1 $\gamma = 1$ — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	Adj+D-CBG (first-order)	0.929 (+0.617)	0.542 (−0.454)	0.515 (+0.203)	0.357 (+0.082)	0.407 (+0.118)
	Adj+D-CBG (exact)	0.924 (+0.612)	0.550 (−0.446)	0.522 (+0.210)	0.361 (+0.086)	0.412 (+0.123)
	Adj+IW	0.925 (+0.613)	0.573 (−0.423)	0.542 (+0.230)	0.361 (+0.086)	0.415 (+0.126)
2	Adj+D-CBG (first-order)	0.867 (+0.599)	0.651 (−0.312)	0.548 (+0.286)	0.397 (+0.153)	0.481 (+0.226)
	Adj+D-CBG (exact)	0.868 (+0.600)	0.658 (−0.305)	0.558 (+0.296)	0.404 (+0.160)	0.489 (+0.235)
	Adj+IW	0.867 (+0.599)	0.678 (−0.285)	0.577 (+0.315)	0.440 (+0.196)	0.515 (+0.260)
4	Adj+D-CBG (first-order)	0.768 (+0.475)	0.732 (−0.244)	0.553 (+0.268)	0.409 (+0.147)	0.483 (+0.208)
	Adj+D-CBG (exact)	0.772 (+0.479)	0.736 (−0.240)	0.558 (+0.273)	0.423 (+0.161)	0.494 (+0.219)
	Adj+IW	0.765 (+0.472)	0.732 (−0.244)	0.556 (+0.271)	0.434 (+0.172)	0.503 (+0.228)
8	Adj+D-CBG (first-order)	0.761 (+0.489)	0.671 (−0.293)	0.478 (+0.222)	0.367 (+0.135)	0.458 (+0.209)
	Adj+D-CBG (exact)	0.763 (+0.491)	0.677 (−0.287)	0.483 (+0.227)	0.372 (+0.140)	0.463 (+0.215)
	Adj+IW	0.760 (+0.488)	0.685 (−0.279)	0.487 (+0.231)	0.392 (+0.160)	0.480 (+0.232)
16	Adj+D-CBG (first-order)	0.749 (+0.490)	0.647 (−0.283)	0.448 (+0.208)	0.346 (+0.125)	0.449 (+0.212)
	Adj+D-CBG (exact)	0.760 (+0.501)	0.650 (−0.280)	0.459 (+0.219)	0.349 (+0.128)	0.456 (+0.218)
	Adj+IW	0.747 (+0.488)	0.653 (−0.277)	0.447 (+0.207)	0.389 (+0.168)	0.485 (+0.247)
32	Adj+D-CBG (first-order)	0.721 (+0.470)	0.633 (−0.265)	0.428 (+0.207)	0.336 (+0.134)	0.435 (+0.214)
	Adj+D-CBG (exact)	0.727 (+0.476)	0.643 (−0.255)	0.441 (+0.220)	0.344 (+0.142)	0.443 (+0.222)
	Adj+IW	0.723 (+0.472)	0.635 (−0.263)	0.427 (+0.206)	0.376 (+0.174)	0.468 (+0.247)
64	Adj+D-CBG (first-order)	0.822 (+0.587)	0.523 (−0.378)	0.395 (+0.193)	0.302 (+0.124)	0.428 (+0.229)
	Adj+D-CBG (exact)	0.822 (+0.587)	0.527 (−0.374)	0.399 (+0.197)	0.313 (+0.135)	0.438 (+0.239)
	Adj+IW	0.828 (+0.593)	0.511 (−0.390)	0.393 (+0.191)	0.359 (+0.181)	0.474 (+0.275)

5.9.2 $\gamma = 1$ — unseen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	Adj+D-CBG (first-order)	0.929 (+0.617)	0.540 (−0.456)	0.513 (+0.201)	0.357 (+0.082)	0.407 (+0.118)
	Adj+D-CBG (exact)	0.929 (+0.617)	0.541 (−0.455)	0.514 (+0.202)	0.357 (+0.082)	0.408 (+0.119)
	Adj+IW	0.928 (+0.616)	0.556 (−0.440)	0.527 (+0.215)	0.354 (+0.079)	0.406 (+0.117)
2	Adj+D-CBG (first-order)	0.865 (+0.597)	0.649 (−0.314)	0.546 (+0.284)	0.395 (+0.151)	0.479 (+0.224)
	Adj+D-CBG (exact)	0.868 (+0.600)	0.651 (−0.312)	0.549 (+0.287)	0.397 (+0.153)	0.481 (+0.227)
	Adj+IW	0.863 (+0.595)	0.677 (−0.286)	0.574 (+0.312)	0.431 (+0.187)	0.507 (+0.252)
4	Adj+D-CBG (first-order)	0.768 (+0.475)	0.734 (−0.242)	0.555 (+0.270)	0.409 (+0.147)	0.482 (+0.207)
	Adj+D-CBG (exact)	0.767 (+0.474)	0.733 (−0.243)	0.553 (+0.268)	0.409 (+0.147)	0.483 (+0.208)
	Adj+IW	0.765 (+0.472)	0.738 (−0.238)	0.561 (+0.276)	0.439 (+0.177)	0.506 (+0.231)
8	Adj+D-CBG (first-order)	0.761 (+0.489)	0.669 (−0.295)	0.476 (+0.220)	0.368 (+0.136)	0.458 (+0.209)
	Adj+D-CBG (exact)	0.761 (+0.489)	0.669 (−0.295)	0.476 (+0.220)	0.367 (+0.135)	0.457 (+0.209)
	Adj+IW	0.758 (+0.486)	0.690 (−0.274)	0.490 (+0.234)	0.393 (+0.161)	0.481 (+0.232)
16	Adj+D-CBG (first-order)	0.749 (+0.490)	0.647 (−0.283)	0.448 (+0.208)	0.349 (+0.128)	0.451 (+0.214)
	Adj+D-CBG (exact)	0.749 (+0.490)	0.648 (−0.282)	0.449 (+0.209)	0.349 (+0.128)	0.451 (+0.213)
	Adj+IW	0.748 (+0.489)	0.653 (−0.277)	0.448 (+0.208)	0.389 (+0.168)	0.485 (+0.247)
32	Adj+D-CBG (first-order)	0.720 (+0.469)	0.632 (−0.266)	0.426 (+0.205)	0.335 (+0.133)	0.434 (+0.213)
	Adj+D-CBG (exact)	0.721 (+0.470)	0.633 (−0.265)	0.428 (+0.207)	0.337 (+0.135)	0.436 (+0.216)
	Adj+IW	0.723 (+0.472)	0.633 (−0.265)	0.425 (+0.204)	0.378 (+0.176)	0.469 (+0.248)
64	Adj+D-CBG (first-order)	0.821 (+0.586)	0.525 (−0.376)	0.395 (+0.193)	0.301 (+0.123)	0.428 (+0.229)
	Adj+D-CBG (exact)	0.822 (+0.587)	0.523 (−0.378)	0.395 (+0.193)	0.309 (+0.131)	0.432 (+0.233)
	Adj+IW	0.828 (+0.593)	0.510 (−0.391)	0.392 (+0.190)	0.359 (+0.181)	0.474 (+0.275)

5.9.3 $\gamma = 4$ — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	Adj+D-CBG (first-order)	0.930 (+0.618)	0.549 (−0.447)	0.522 (+0.210)	0.358 (+0.083)	0.408 (+0.119)
	Adj+D-CBG (exact)	0.921 (+0.609)	0.574 (−0.422)	0.542 (+0.230)	0.376 (+0.101)	0.429 (+0.140)
	Adj+IW	0.925 (+0.613)	0.573 (−0.423)	0.542 (+0.230)	0.361 (+0.086)	0.415 (+0.126)
2	Adj+D-CBG (first-order)	0.868 (+0.600)	0.656 (−0.307)	0.555 (+0.293)	0.400 (+0.156)	0.484 (+0.229)
	Adj+D-CBG (exact)	0.873 (+0.605)	0.680 (−0.283)	0.584 (+0.322)	0.412 (+0.168)	0.499 (+0.245)
	Adj+IW	0.867 (+0.599)	0.678 (−0.285)	0.577 (+0.315)	0.440 (+0.196)	0.515 (+0.260)
4	Adj+D-CBG (first-order)	0.768 (+0.475)	0.733 (−0.243)	0.555 (+0.270)	0.411 (+0.149)	0.485 (+0.210)
	Adj+D-CBG (exact)	0.785 (+0.492)	0.733 (−0.243)	0.571 (+0.286)	0.438 (+0.176)	0.515 (+0.240)
	Adj+IW	0.765 (+0.472)	0.732 (−0.244)	0.556 (+0.271)	0.434 (+0.172)	0.503 (+0.228)
8	Adj+D-CBG (first-order)	0.763 (+0.491)	0.671 (−0.293)	0.480 (+0.224)	0.372 (+0.140)	0.462 (+0.213)
	Adj+D-CBG (exact)	0.763 (+0.491)	0.681 (−0.283)	0.491 (+0.235)	0.383 (+0.151)	0.473 (+0.224)
	Adj+IW	0.760 (+0.488)	0.685 (−0.279)	0.487 (+0.231)	0.392 (+0.160)	0.480 (+0.232)
16	Adj+D-CBG (first-order)	0.751 (+0.492)	0.649 (−0.281)	0.451 (+0.211)	0.346 (+0.125)	0.451 (+0.213)
	Adj+D-CBG (exact)	0.768 (+0.509)	0.651 (−0.279)	0.466 (+0.226)	0.370 (+0.149)	0.474 (+0.236)
	Adj+IW	0.747 (+0.488)	0.653 (−0.277)	0.447 (+0.207)	0.389 (+0.168)	0.485 (+0.247)
32	Adj+D-CBG (first-order)	0.725 (+0.474)	0.631 (−0.267)	0.431 (+0.210)	0.335 (+0.133)	0.436 (+0.215)
	Adj+D-CBG (exact)	0.744 (+0.493)	0.653 (−0.245)	0.462 (+0.241)	0.362 (+0.160)	0.465 (+0.245)
	Adj+IW	0.723 (+0.472)	0.635 (−0.263)	0.427 (+0.206)	0.376 (+0.174)	0.468 (+0.247)
64	Adj+D-CBG (first-order)	0.823 (+0.588)	0.524 (−0.377)	0.397 (+0.195)	0.303 (+0.125)	0.429 (+0.230)
	Adj+D-CBG (exact)	0.830 (+0.595)	0.532 (−0.369)	0.412 (+0.210)	0.336 (+0.158)	0.459 (+0.259)
	Adj+IW	0.828 (+0.593)	0.511 (−0.390)	0.393 (+0.191)	0.359 (+0.181)	0.474 (+0.275)

5.9.4 $\gamma = 4$ — unseen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	Adj+D-CBG (first-order)	0.928 (+0.616)	0.543 (−0.453)	0.516 (+0.204)	0.356 (+0.081)	0.407 (+0.118)
	Adj+D-CBG (exact)	0.927 (+0.615)	0.555 (−0.441)	0.527 (+0.215)	0.361 (+0.086)	0.415 (+0.126)
	Adj+IW	0.928 (+0.616)	0.556 (−0.440)	0.527 (+0.215)	0.354 (+0.079)	0.406 (+0.117)
2	Adj+D-CBG (first-order)	0.865 (+0.597)	0.651 (−0.312)	0.548 (+0.286)	0.392 (+0.148)	0.476 (+0.222)
	Adj+D-CBG (exact)	0.871 (+0.603)	0.655 (−0.308)	0.554 (+0.292)	0.399 (+0.155)	0.484 (+0.229)
	Adj+IW	0.863 (+0.595)	0.677 (−0.286)	0.574 (+0.312)	0.431 (+0.187)	0.507 (+0.252)
4	Adj+D-CBG (first-order)	0.767 (+0.474)	0.729 (−0.247)	0.550 (+0.265)	0.404 (+0.142)	0.479 (+0.204)
	Adj+D-CBG (exact)	0.768 (+0.475)	0.729 (−0.247)	0.553 (+0.268)	0.415 (+0.153)	0.486 (+0.212)
	Adj+IW	0.765 (+0.472)	0.738 (−0.238)	0.561 (+0.276)	0.439 (+0.177)	0.506 (+0.231)
8	Adj+D-CBG (first-order)	0.764 (+0.492)	0.674 (−0.290)	0.482 (+0.226)	0.373 (+0.141)	0.462 (+0.213)
	Adj+D-CBG (exact)	0.763 (+0.491)	0.670 (−0.294)	0.480 (+0.224)	0.371 (+0.139)	0.460 (+0.212)
	Adj+IW	0.758 (+0.486)	0.690 (−0.274)	0.490 (+0.234)	0.393 (+0.161)	0.481 (+0.232)
16	Adj+D-CBG (first-order)	0.749 (+0.490)	0.645 (−0.285)	0.447 (+0.207)	0.347 (+0.126)	0.449 (+0.212)
	Adj+D-CBG (exact)	0.752 (+0.493)	0.644 (−0.286)	0.449 (+0.209)	0.355 (+0.134)	0.455 (+0.218)
	Adj+IW	0.748 (+0.489)	0.653 (−0.277)	0.448 (+0.208)	0.389 (+0.168)	0.485 (+0.247)
32	Adj+D-CBG (first-order)	0.719 (+0.468)	0.636 (−0.262)	0.428 (+0.207)	0.337 (+0.135)	0.435 (+0.215)
	Adj+D-CBG (exact)	0.721 (+0.470)	0.638 (−0.260)	0.431 (+0.210)	0.344 (+0.142)	0.441 (+0.220)
	Adj+IW	0.723 (+0.472)	0.633 (−0.265)	0.425 (+0.204)	0.378 (+0.176)	0.469 (+0.248)
64	Adj+D-CBG (first-order)	0.820 (+0.585)	0.525 (−0.376)	0.395 (+0.193)	0.302 (+0.124)	0.429 (+0.230)
	Adj+D-CBG (exact)	0.824 (+0.589)	0.516 (−0.385)	0.391 (+0.189)	0.307 (+0.129)	0.432 (+0.233)
	Adj+IW	0.828 (+0.593)	0.510 (−0.391)	0.392 (+0.190)	0.359 (+0.181)	0.474 (+0.275)

- **The mask carries this arm too, and §5.5's division of labour survives the scorer swap.** On top of adj-only, exact D-CBG at $\gamma = 1$ adds +0.003 to +0.014 $v_{\wedge a}$ and +0.005 to +0.014 EM; the first-order variant adds −0.002 to +0.002 — inert on top of the mask, exactly as it is without it.
- **At $\gamma = 4$ exact Adj+D-CBG reaches Adj+IW on $v_{\wedge a}$ and passes it at large blocks (seen):** $\Delta v_{\wedge a}$ vs Adj+IW runs 0.000 to +0.035, positive from blk16 up (blk32 +0.035, against the ± 0.012 floor). The three-seed replication of §5.10 tempers this: the blk16 lead survives the spread, the blk32/64 lead does not. But Adj+IW keeps the **EM** lead at almost every block (−0.028 to +0.015 for Adj+D-CBG) — the classifier's tilt converts more reveals into surviving legal paths, while the discriminator's one-ply lookahead still picks the *right* legal paths. With the legality mask in place the two scorers are buying different things.
- **The γ boost does not transfer zero-shot.** The seen $\gamma = 4$ gain over $\gamma = 1$ is +0.007 to +0.026; unseen it collapses to −0.004 to +0.013. Adj+IW's seen/unseen parity (§3.3) remains the differentiator that γ cannot buy.
- **Read with the usual caveats:** γ was tuned for the baseline only (Adj+IW stays at $w_{\gamma} = 1$), every cell is a single run against the ± 0.012 floor of §2.2, and the classifier and discriminator are trained on the same pools — so this is the like-for-like mechanism comparison, now at matched *sampler* as well as matched data.

5.10 Three-seed replication of the headline arms

Every earlier cell is a single run; this table replicates the fam 0.05 / l2r / seen headline arms over sampling seeds {7, 8, 9} — the same 1,000 reserved except_0 OD pairs each time, only the generator seed varies, so the spread is the sampler's own variance on the fixed benchmark (this partially discharges E3). The 1,000-pair set is a protocol

choice, not a data limit: except_0 holds 1,930,710 OD paths. Cells: mean $\pm$ sample std (n = 3); seed 7 is the run reported in §§3–5.9. Bold: best mean in the block. IW-only was replicated as well and is in the JSON (`sets_res/seed_stats_f005.json`), omitted here for width.

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	base	0.307 $\pm$ 0.006	0.995 $\pm$ 0.002	0.306 $\pm$ 0.006	0.271 $\pm$ 0.004	0.285 $\pm$ 0.004
	Adj-only	0.926 $\pm$ 0.003	0.550 $\pm$ 0.011	0.521 $\pm$ 0.009	0.361 $\pm$ 0.005	0.412 $\pm$ 0.006
	Adj+IW	0.924 $\pm$ 0.002	0.571 $\pm$ 0.002	0.541 $\pm$ 0.002	0.369 $\pm$ 0.007	0.422 $\pm$ 0.006
	Adj+D-CBG (exact, γ =1)	0.923 $\pm$ 0.002	0.557 $\pm$ 0.008	0.528 $\pm$ 0.007	0.366 $\pm$ 0.004	0.417 $\pm$ 0.005
	Adj+D-CBG (exact, γ =4)	0.919 $\pm$ 0.002	0.577 $\pm$ 0.008	0.544 $\pm$ 0.008	0.375 $\pm$ 0.005	0.429 $\pm$ 0.001
2	base	0.276 $\pm$ 0.007	0.964 $\pm$ 0.001	0.269 $\pm$ 0.006	0.253 $\pm$ 0.010	0.264 $\pm$ 0.008
	Adj-only	0.864 $\pm$ 0.006	0.659 $\pm$ 0.008	0.556 $\pm$ 0.011	0.397 $\pm$ 0.008	0.476 $\pm$ 0.009
	Adj+IW	0.866 $\pm$ 0.004	0.675 $\pm$ 0.005	0.574 $\pm$ 0.006	0.436 $\pm$ 0.006	0.512 $\pm$ 0.006
	Adj+D-CBG (exact, γ =1)	0.866 $\pm$ 0.006	0.666 $\pm$ 0.007	0.565 $\pm$ 0.009	0.406 $\pm$ 0.011	0.486 $\pm$ 0.011
	Adj+D-CBG (exact, γ =4)	0.869 $\pm$ 0.006	0.686 $\pm$ 0.005	0.587 $\pm$ 0.008	0.417 $\pm$ 0.011	0.499 $\pm$ 0.011
4	base	0.299 $\pm$ 0.007	0.976 $\pm$ 0.001	0.291 $\pm$ 0.006	0.267 $\pm$ 0.008	0.281 $\pm$ 0.008
	Adj-only	0.765 $\pm$ 0.004	0.729 $\pm$ 0.011	0.550 $\pm$ 0.011	0.408 $\pm$ 0.004	0.482 $\pm$ 0.004
	Adj+IW	0.768 $\pm$ 0.003	0.727 $\pm$ 0.007	0.552 $\pm$ 0.008	0.429 $\pm$ 0.005	0.500 $\pm$ 0.003
	Adj+D-CBG (exact, γ =1)	0.769 $\pm$ 0.003	0.731 $\pm$ 0.011	0.554 $\pm$ 0.009	0.418 $\pm$ 0.004	0.491 $\pm$ 0.003
	Adj+D-CBG (exact, γ =4)	0.779 $\pm$ 0.006	0.736 $\pm$ 0.011	0.568 $\pm$ 0.009	0.427 $\pm$ 0.010	0.506 $\pm$ 0.008
8	base	0.270 $\pm$ 0.005	0.966 $\pm$ 0.002	0.258 $\pm$ 0.003	0.239 $\pm$ 0.008	0.252 $\pm$ 0.005
	Adj-only	0.763 $\pm$ 0.003	0.670 $\pm$ 0.001	0.474 $\pm$ 0.002	0.363 $\pm$ 0.003	0.454 $\pm$ 0.002
	Adj+IW	0.758 $\pm$ 0.005	0.693 $\pm$ 0.007	0.492 $\pm$ 0.004	0.398 $\pm$ 0.010	0.484 $\pm$ 0.008
	Adj+D-CBG (exact, γ =1)	0.764 $\pm$ 0.002	0.675 $\pm$ 0.002	0.481 $\pm$ 0.003	0.370 $\pm$ 0.003	0.459 $\pm$ 0.004
	Adj+D-CBG (exact, γ =4)	0.768 $\pm$ 0.006	0.684 $\pm$ 0.003	0.495 $\pm$ 0.003	0.379 $\pm$ 0.009	0.470 $\pm$ 0.003
16	base	0.261 $\pm$ 0.007	0.942 $\pm$ 0.011	0.244 $\pm$ 0.008	0.223 $\pm$ 0.007	0.239 $\pm$ 0.007
	Adj-only	0.753 $\pm$ 0.011	0.642 $\pm$ 0.008	0.442 $\pm$ 0.015	0.342 $\pm$ 0.005	0.446 $\pm$ 0.002
	Adj+IW	0.751 $\pm$ 0.004	0.647 $\pm$ 0.008	0.446 $\pm$ 0.005	0.389 $\pm$ 0.010	0.487 $\pm$ 0.006
	Adj+D-CBG (exact, γ =1)	0.759 $\pm$ 0.007	0.647 $\pm$ 0.007	0.453 $\pm$ 0.012	0.349 $\pm$ 0.003	0.455 $\pm$ 0.002
	Adj+D-CBG (exact, γ =4)	0.778 $\pm$ 0.012	0.646 $\pm$ 0.005	0.468 $\pm$ 0.006	0.372 $\pm$ 0.002	0.478 $\pm$ 0.005
32	base	0.261 $\pm$ 0.011	0.902 $\pm$ 0.008	0.229 $\pm$ 0.011	0.209 $\pm$ 0.007	0.230 $\pm$ 0.009
	Adj-only	0.716 $\pm$ 0.012	0.633 $\pm$ 0.015	0.419 $\pm$ 0.010	0.336 $\pm$ 0.004	0.435 $\pm$ 0.005
	Adj+IW	0.723 $\pm$ 0.003	0.648 $\pm$ 0.012	0.435 $\pm$ 0.007	0.374 $\pm$ 0.009	0.466 $\pm$ 0.006
	Adj+D-CBG (exact, γ =1)	0.721 $\pm$ 0.007	0.640 $\pm$ 0.009	0.429 $\pm$ 0.013	0.344 $\pm$ 0.003	0.443 $\pm$ 0.003
	Adj+D-CBG (exact, γ =4)	0.739 $\pm$ 0.004	0.647 $\pm$ 0.011	0.449 $\pm$ 0.015	0.365 $\pm$ 0.003	0.467 $\pm$ 0.002
64	base	0.234 $\pm$ 0.004	0.901 $\pm$ 0.002	0.206 $\pm$ 0.003	0.182 $\pm$ 0.004	0.201 $\pm$ 0.002
	Adj-only	0.829 $\pm$ 0.008	0.513 $\pm$ 0.012	0.392 $\pm$ 0.009	0.308 $\pm$ 0.006	0.433 $\pm$ 0.004
	Adj+IW	0.827 $\pm$ 0.006	0.529 $\pm$ 0.021	0.409 $\pm$ 0.018	0.361 $\pm$ 0.009	0.476 $\pm$ 0.004
	Adj+D-CBG (exact, γ =1)	0.832 $\pm$ 0.009	0.519 $\pm$ 0.010	0.399 $\pm$ 0.011	0.319 $\pm$ 0.005	0.443 $\pm$ 0.005
	Adj+D-CBG (exact, γ =4)	0.832 $\pm$ 0.002	0.528 $\pm$ 0.004	0.410 $\pm$ 0.005	0.335 $\pm$ 0.001	0.459 $\pm$ 0.001

- The single-run noise floor was honest.** Across all arms and blocks the v_{aa} std is 0.007 at the median and 0.018 at the worst cell — bracketing the ± 0.012 floor that every earlier section read its gaps against.
- What survives replication is the EM split, and what softens is the v_{aa} overtake.** Adj+IW leads Adj+D-CBG ($\gamma = 4$) on EM at every block ≥ 2 by 0.002–0.026, well beyond the spread — the discriminator's pick-the-right-legal-path advantage is a property, not a seed. On v_{aa} , Adj+D-CBG ($\gamma = 4$) is never behind (gaps +0.001 to +0.022), but the gap clears the combined spread only at blk16 (+0.022); the seed-7 blk32/64 overtake of §5.9 shrinks to +0.013 / +0.001 under the 3-seed mean — partly seed luck, which is exactly what this table exists to catch.
- The classifier's increment over the bare mask is small but directionally consistent:** exact D-CBG at $\gamma = 1$ adds +0.004 to +0.011 v_{aa} over adj-only, positive at all seven blocks (7/7 sign agreement across three seeds).

5.11 Final summary tables

The mask-free guidance comparison in four tables — {seen, unseen} $\times \gamma \in \{1, 4\}$ — in the §5.7 layout: D-CBG in both variants against IW-only, no adjacency restriction anywhere. Collected from §§3, 5.7–5.8; no new runs — every number appears earlier in this document. Fam 0.05, l2r, the 1,000 reserved except_0 pairs, RAW, seed 7 throughout. Parentheses: gain over that block's unguided base; bold: best of the three arms in the block. The IW rows always use $w_\gamma = 1$ — γ was swept for D-CBG only, so the $\gamma = 4$ tables repeat the same IW rows as the $\gamma = 1$ tables.

5.11.1 $\gamma = 1$ — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.314 (+0.002)	0.996 (0.000)	0.314 (+0.002)	0.278 (+0.003)	0.291 (+0.002)
	D-CBG (exact)	0.327 (+0.015)	0.994 (−0.002)	0.325 (+0.013)	0.287 (+0.012)	0.301 (+0.012)
	IW-only	0.331 (+0.019)	0.992 (−0.004)	0.328 (+0.016)	0.287 (+0.012)	0.303 (+0.014)
2	D-CBG (first-order)	0.266 (−0.002)	0.963 (0.000)	0.260 (−0.002)	0.243 (−0.001)	0.253 (−0.002)
	D-CBG (exact)	0.308 (+0.040)	0.962 (−0.001)	0.297 (+0.035)	0.274 (+0.030)	0.288 (+0.033)
	IW-only	0.322 (+0.054)	0.963 (0.000)	0.317 (+0.055)	0.292 (+0.048)	0.305 (+0.050)
4	D-CBG (first-order)	0.294 (+0.001)	0.977 (+0.001)	0.286 (+0.001)	0.265 (+0.003)	0.277 (+0.002)
	D-CBG (exact)	0.319 (+0.026)	0.975 (−0.001)	0.306 (+0.021)	0.279 (+0.017)	0.295 (+0.020)
	IW-only	0.337 (+0.044)	0.965 (−0.011)	0.320 (+0.035)	0.305 (+0.043)	0.318 (+0.043)
8	D-CBG (first-order)	0.271 (−0.001)	0.964 (0.000)	0.255 (−0.001)	0.231 (−0.001)	0.247 (−0.001)
	D-CBG (exact)	0.307 (+0.035)	0.958 (−0.006)	0.283 (+0.027)	0.256 (+0.024)	0.278 (+0.029)
	IW-only	0.321 (+0.049)	0.963 (−0.001)	0.303 (+0.047)	0.289 (+0.057)	0.304 (+0.055)
16	D-CBG (first-order)	0.259 (0.000)	0.930 (0.000)	0.240 (0.000)	0.221 (0.000)	0.238 (0.000)
	D-CBG (exact)	0.315 (+0.056)	0.933 (+0.003)	0.281 (+0.041)	0.250 (+0.029)	0.275 (+0.038)
	IW-only	0.312 (+0.053)	0.934 (+0.004)	0.281 (+0.041)	0.259 (+0.038)	0.280 (+0.043)
32	D-CBG (first-order)	0.251 (0.000)	0.899 (+0.001)	0.222 (+0.001)	0.202 (0.000)	0.221 (0.000)
	D-CBG (exact)	0.317 (+0.066)	0.890 (−0.008)	0.269 (+0.048)	0.237 (+0.035)	0.265 (+0.044)
	IW-only	0.322 (+0.071)	0.894 (−0.004)	0.267 (+0.046)	0.259 (+0.057)	0.283 (+0.062)
64	D-CBG (first-order)	0.236 (+0.001)	0.901 (0.000)	0.203 (+0.001)	0.179 (+0.001)	0.200 (+0.001)
	D-CBG (exact)	0.301 (+0.066)	0.890 (−0.011)	0.246 (+0.044)	0.210 (+0.032)	0.244 (+0.045)
	IW-only	0.307 (+0.072)	0.879 (−0.022)	0.247 (+0.045)	0.251 (+0.073)	0.271 (+0.072)

5.11.2 $\gamma = 1$ — unseen (except 1–99 → except_0)

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.312 (0.000)	0.997 (+0.001)	0.312 (0.000)	0.275 (0.000)	0.289 (0.000)
	D-CBG (exact)	0.327 (+0.015)	0.994 (−0.002)	0.325 (+0.013)	0.284 (+0.009)	0.300 (+0.011)
	IW-only	0.327 (+0.015)	0.992 (−0.004)	0.324 (+0.012)	0.283 (+0.008)	0.299 (+0.010)
2	D-CBG (first-order)	0.269 (+0.001)	0.963 (0.000)	0.263 (+0.001)	0.245 (+0.001)	0.256 (+0.001)
	D-CBG (exact)	0.310 (+0.042)	0.962 (−0.001)	0.298 (+0.036)	0.275 (+0.031)	0.289 (+0.035)
	IW-only	0.328 (+0.060)	0.966 (+0.003)	0.321 (+0.059)	0.298 (+0.054)	0.311 (+0.057)
4	D-CBG (first-order)	0.290 (−0.003)	0.976 (0.000)	0.282 (−0.003)	0.263 (+0.001)	0.275 (0.000)
	D-CBG (exact)	0.328 (+0.035)	0.974 (−0.002)	0.316 (+0.031)	0.283 (+0.021)	0.301 (+0.026)
	IW-only	0.336 (+0.043)	0.972 (−0.004)	0.326 (+0.041)	0.303 (+0.041)	0.316 (+0.041)
8	D-CBG (first-order)	0.272 (0.000)	0.964 (0.000)	0.256 (0.000)	0.232 (0.000)	0.248 (0.000)
	D-CBG (exact)	0.314 (+0.042)	0.959 (−0.005)	0.288 (+0.032)	0.254 (+0.022)	0.278 (+0.030)
	IW-only	0.323 (+0.051)	0.963 (−0.001)	0.304 (+0.048)	0.290 (+0.058)	0.305 (+0.056)
16	D-CBG (first-order)	0.260 (+0.001)	0.931 (+0.001)	0.241 (+0.001)	0.222 (+0.001)	0.239 (+0.001)
	D-CBG (exact)	0.312 (+0.053)	0.933 (+0.003)	0.278 (+0.038)	0.249 (+0.028)	0.273 (+0.036)
	IW-only	0.315 (+0.056)	0.934 (+0.004)	0.283 (+0.043)	0.263 (+0.042)	0.284 (+0.046)
32	D-CBG (first-order)	0.250 (−0.001)	0.897 (−0.001)	0.220 (−0.001)	0.201 (−0.001)	0.220 (−0.001)
	D-CBG (exact)	0.314 (+0.063)	0.885 (−0.013)	0.262 (+0.041)	0.233 (+0.031)	0.262 (+0.042)
	IW-only	0.322 (+0.071)	0.896 (−0.002)	0.269 (+0.048)	0.263 (+0.061)	0.285 (+0.065)
64	D-CBG (first-order)	0.234 (−0.001)	0.901 (0.000)	0.201 (−0.001)	0.177 (−0.001)	0.198 (−0.001)
	D-CBG (exact)	0.299 (+0.064)	0.896 (−0.005)	0.245 (+0.043)	0.210 (+0.032)	0.243 (+0.044)
	IW-only	0.306 (+0.071)	0.879 (−0.022)	0.248 (+0.046)	0.250 (+0.072)	0.270 (+0.071)

5.11.3 $\gamma = 4$ — seen

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.313 (+0.001)	0.996 (0.000)	0.313 (+0.001)	0.278 (+0.003)	0.291 (+0.002)
	D-CBG (exact)	0.388 (+0.076)	0.981 (−0.015)	0.384 (+0.072)	0.317 (+0.042)	0.341 (+0.052)
	IW-only	0.331 (+0.019)	0.992 (−0.004)	0.328 (+0.016)	0.287 (+0.012)	0.303 (+0.014)
2	D-CBG (first-order)	0.268 (0.000)	0.965 (+0.002)	0.264 (+0.002)	0.247 (+0.003)	0.256 (+0.002)
	D-CBG (exact)	0.398 (+0.130)	0.951 (−0.012)	0.372 (+0.110)	0.326 (+0.082)	0.352 (+0.097)
	IW-only	0.322 (+0.054)	0.963 (0.000)	0.317 (+0.055)	0.292 (+0.048)	0.305 (+0.050)
4	D-CBG (first-order)	0.296 (+0.003)	0.975 (−0.001)	0.287 (+0.002)	0.264 (+0.002)	0.277 (+0.002)
	D-CBG (exact)	0.398 (+0.105)	0.950 (−0.026)	0.363 (+0.078)	0.322 (+0.060)	0.351 (+0.076)
	IW-only	0.337 (+0.044)	0.965 (−0.011)	0.320 (+0.035)	0.305 (+0.043)	0.318 (+0.043)
8	D-CBG (first-order)	0.275 (+0.003)	0.961 (−0.003)	0.257 (+0.001)	0.234 (+0.002)	0.251 (+0.003)
	D-CBG (exact)	0.398 (+0.126)	0.917 (−0.047)	0.341 (+0.085)	0.305 (+0.073)	0.340 (+0.092)
	IW-only	0.321 (+0.049)	0.963 (−0.001)	0.303 (+0.047)	0.289 (+0.057)	0.304 (+0.055)
16	D-CBG (first-order)	0.260 (+0.001)	0.928 (−0.002)	0.240 (0.000)	0.222 (+0.001)	0.239 (+0.001)
	D-CBG (exact)	0.462 (+0.203)	0.876 (−0.054)	0.355 (+0.115)	0.315 (+0.094)	0.365 (+0.128)
	IW-only	0.312 (+0.053)	0.934 (+0.004)	0.281 (+0.041)	0.259 (+0.038)	0.280 (+0.043)
32	D-CBG (first-order)	0.254 (+0.003)	0.897 (−0.001)	0.223 (+0.002)	0.203 (+0.001)	0.222 (+0.002)
	D-CBG (exact)	0.485 (+0.234)	0.826 (−0.072)	0.356 (+0.135)	0.306 (+0.104)	0.364 (+0.143)
	IW-only	0.322 (+0.071)	0.894 (−0.004)	0.267 (+0.046)	0.259 (+0.057)	0.283 (+0.062)
64	D-CBG (first-order)	0.235 (0.000)	0.902 (+0.001)	0.202 (0.000)	0.179 (+0.001)	0.200 (+0.001)
	D-CBG (exact)	0.506 (+0.271)	0.790 (−0.111)	0.345 (+0.143)	0.290 (+0.112)	0.360 (+0.161)
	IW-only	0.307 (+0.072)	0.879 (−0.022)	0.247 (+0.045)	0.251 (+0.073)	0.271 (+0.072)

5.11.4 $\gamma = 4$ — unseen (except 1–99 → except_0)

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order)	0.314 (+0.002)	0.997 (+0.001)	0.314 (+0.002)	0.275 (0.000)	0.290 (+0.001)
	D-CBG (exact)	0.393 (+0.081)	0.980 (−0.016)	0.387 (+0.075)	0.317 (+0.042)	0.340 (+0.051)
	IW-only	0.327 (+0.015)	0.992 (−0.004)	0.324 (+0.012)	0.283 (+0.008)	0.299 (+0.010)
2	D-CBG (first-order)	0.268 (0.000)	0.963 (0.000)	0.262 (0.000)	0.245 (+0.001)	0.255 (+0.001)
	D-CBG (exact)	0.397 (+0.129)	0.947 (−0.016)	0.365 (+0.103)	0.315 (+0.071)	0.343 (+0.088)
	IW-only	0.328 (+0.060)	0.966 (+0.003)	0.321 (+0.059)	0.298 (+0.054)	0.311 (+0.057)
4	D-CBG (first-order)	0.290 (−0.003)	0.976 (0.000)	0.282 (−0.003)	0.263 (+0.001)	0.275 (0.000)
	D-CBG (exact)	0.403 (+0.110)	0.945 (−0.031)	0.366 (+0.081)	0.316 (+0.054)	0.348 (+0.073)
	IW-only	0.336 (+0.043)	0.972 (−0.004)	0.326 (+0.041)	0.303 (+0.041)	0.316 (+0.041)
8	D-CBG (first-order)	0.270 (−0.002)	0.963 (−0.001)	0.254 (−0.002)	0.230 (−0.002)	0.246 (−0.002)
	D-CBG (exact)	0.441 (+0.169)	0.910 (−0.054)	0.367 (+0.111)	0.316 (+0.084)	0.359 (+0.111)
	IW-only	0.323 (+0.051)	0.963 (−0.001)	0.304 (+0.048)	0.290 (+0.058)	0.305 (+0.056)
16	D-CBG (first-order)	0.262 (+0.003)	0.932 (+0.002)	0.242 (+0.002)	0.224 (+0.003)	0.241 (+0.003)
	D-CBG (exact)	0.443 (+0.184)	0.867 (−0.063)	0.336 (+0.096)	0.297 (+0.076)	0.345 (+0.108)
	IW-only	0.315 (+0.056)	0.934 (+0.004)	0.283 (+0.043)	0.263 (+0.042)	0.284 (+0.046)
32	D-CBG (first-order)	0.251 (0.000)	0.896 (−0.002)	0.221 (0.000)	0.202 (0.000)	0.221 (0.000)
	D-CBG (exact)	0.466 (+0.215)	0.823 (−0.075)	0.327 (+0.106)	0.297 (+0.095)	0.352 (+0.131)
	IW-only	0.322 (+0.071)	0.896 (−0.002)	0.269 (+0.048)	0.263 (+0.061)	0.285 (+0.065)
64	D-CBG (first-order)	0.235 (0.000)	0.902 (+0.001)	0.202 (0.000)	0.178 (0.000)	0.200 (0.000)
	D-CBG (exact)	0.526 (+0.291)	0.768 (−0.133)	0.336 (+0.134)	0.279 (+0.101)	0.356 (+0.157)
	IW-only	0.306 (+0.071)	0.879 (−0.022)	0.248 (+0.046)	0.250 (+0.072)	0.270 (+0.071)

6. Experiments still to run before submission

Ordered by how much a reviewer's objection would cost us if unanswered.

E1 — Compute-matched best-of-n, and discriminator reranking (highest priority)

The acceptance criterion our guidance targets — every edge legal in A_{scn} , endpoint equal to the destination — is **checkable at test time from exactly the inputs the discriminator receives**. So the first question will be: why not draw n unguided samples and keep one that passes? That is rejection sampling from $p_\theta(\cdot \mid \text{accepted})$, truncated to a budget. The fair n is the cost ratio, which the v4 report §5.6 fixes per block: $n = 18.8$ (blk1), 12.0 (blk2), 6.5 (blk4), 9.4 (blk64) — note it is *largest* at blk1, so the baseline is strongest exactly where AR sits.

A back-of-envelope estimate from the stored per-path records is uncomfortable: at blk4/l2r a single base draw has $v_{\wedge} \approx 0.285$, and among accepted draws 89.5% are exact matches to the reference; if the n draws were independent, best-of-6 would reach $v_{\wedge} \approx 0.866$ and $\text{EM} \approx 0.775$ against adj+IW's 0.556 / 0.434. **The independence assumption is false** — the n draws share an OD pair and a model, so failures are positively correlated and hard pairs fail every time — so the true curve lies below that bound, possibly far below. This must be measured, not computed. Three arms at matched cost:

- ① base best-of- n + hard accept/reject (no learning at all)
- ② base best-of- n **reranked by our discriminator** (the scorer used once, post hoc)
- ③ per-step SNIS guidance (ours)

Report the pass rate as well as the metrics: the gap between the measured pass rate and $1 - (1 - p)^n$ is the correlation, and it is the quantity that decides whether ① is a real threat. If $③ > ② > ①$, the paper gets a clean two-step argument — the discriminator carries signal ($② > ①$), and injecting it per step rather than post hoc is what pays ($③ > ②$) — and §5.3 explains why: ② scores a finished path once, ③ scores a lookahead at every move. Also measure the composition $③ + \text{best-of-}n'$, since the two are not mutually exclusive.

E2 — Retrain the scorers on l2r negatives

Regenerate the unconditional pools with `—order l2r` (a one-line addition to `gen_bd_uncond_pool.py`) and retrain both the IW discriminators and the D-CBG classifiers on them, removing the mismatch disclosed in §2.3. Both methods benefit, so this is not a way of buying an advantage

— it removes an unforced handicap and makes $p_{\text{ref}} = p_0$ as the estimator assumes.

E3 — Seeds on the headline cells

Every cell in §3 is a single run. The blk1 row bounds $v_{\wedge a}$ noise at ≈ 0.012 , so the blk ≥ 8 gains (0.04–0.23) are far outside it, but the blk2/blk4 base and IW-only gains (0.005–0.024) sit at the floor and the headline AR-vs-blk2 gap (+0.035 on $v_{\wedge a}$) is only about three times it. Three seeds on blk1/2/4 $\times$ {base, IW, adj+IW} $\times$ l2r is enough to state that gap with a spread; EM/PC gaps (+0.079 / +0.100) are already safe.

E4 — Instrument the dead-end fallback (cheap, disclosure)

When the Lemma-3 legality mask leaves no candidate with non-negligible probability, the sampler drops the mask for that step (`bd_models.py` , present since the masking was first implemented, so it is in effect for every guided number in this record). This is the right default — it makes adj-only a continuous relaxation of base, agreeing with base exactly where the constraint is inactive or infeasible, whereas emitting END would inject behaviour the model never proposed and dropping the row would bias the eval set. It is also nearly free of consequence for $v_{\wedge a}$, since a fallback path fails validity and an END path fails arrival, and both give $v_{\wedge a} = 0$. What it does affect is *which column* records the failure, and therefore how the validity/arrival trade in §3 reads. Count how often `z  $\leq$  1e-9` fires, per block and per order, and report it next to validity; describe adj-only in the paper as a best-effort constraint with a documented escape hatch rather than a hard one. Add the counter after the queue drains, so the whole record stays on one version of the sampler.

E5 — The metric question, which decides the framing

The reference set is `porto_shrink_SP_*` , i.e. **shortest paths**. Measured by exact match against it, the task is one Dijkstra solves exactly and best-of-n approximates well, and no generative method can win on its own terms. Two ways out, both worth doing: (a) evaluate against **real trajectories** rather than shortest paths, so that the reference is not computable in closed form and a learned model has something to contribute; (b) frame the contribution as **distribution matching** and lead with the metrics rejection sampling cannot game — JSEV over the edge distribution, length calibration, diversity among valid paths — since best-of-n samples $p_0(\cdot \mid \text{accepted})$, which is not q_{exc} . The JSEV machinery already exists (`eval_uncond_jsev.py`). This choice should be settled before the results section is written, because it determines which table is Table 1.

E6 — Ancestral within-block proposals (method extension)

§5.4 identifies the mean-field proposal as what caps the lookahead's usefulness at blk2–4. Sampling the block positions in order, each conditioned on the previous draw through the adjacency mask, would make the lookahead a coherent legal continuation and might extend the benefit to larger blocks. Two caveats before implementing: it costs B sequential multinomial/gather steps per reveal on $(b \cdot n_{\text{is}}, V)$ tensors instead of one batched step — bounded, since the v4 report §5.6 puts discriminator scoring at $\approx 83\%$ of the guided cost at blk4 — and, more importantly, **it changes the proposal distribution**. The current weight $\exp(\gamma \ell)$ is the correct ratio only when the proposal is the denoiser marginal; an ancestral cascade gives each candidate its own product of per-step renormalisers, which does not cancel under self-normalisation. Lemma 3's zero-target-weight argument may extend, but the derivation has to be checked before the code is written, or the estimator is silently biased.

E7 — Second road network

Everything here is Porto, $V = 1,390$, one graph. A workshop track survives that; a main track will ask for a second network. This is the single largest gap between the current record and a main-track submission, and it is also the most expensive item, so it should be decided early rather than late.

E8 — KV caching (optional, wall-clock only)

Not currently implemented anywhere in the codebase. The block-causal structure suits it — committed blocks' K/V never change, so each reveal would recompute only the current block — and it would flatten the block-size dependence of the timing table, whose right arm is dominated by the fact that every reveal re-runs the full canvas. It changes no quality number, and it carries a silent-regression risk (within-block attention is bidirectional, so a completed block's cached K/V are stale until one extra clean forward is run over it). Recommended *after* submission. In the meantime, add a **denoiser-forwards-per-path** row next to the existing scorer-calls row in the timing table, so the implementation-independent cost is visible and the wall-clock caveat is self-evident.

7. Provenance

IW l2r: `run_l2r_v4.sh` $\rightarrow$ `sets_res/va_table_l2r_f005.json` (14 jobs, 2026-07-28 13:34–14:11). IW first-hitting reference: `sets_res/va_table_v4_f005.json` (v4 report §4). D-CBG l2r: `run_l2r_dcbg.sh` $\rightarrow$ `sets_res/dcbg_table_l2r_f005.json` , re-measured 2026-07-29 after the classifier time-conditioning fix (§4; pre-fix tables kept as `*.pre_fix`). D-CBG first-hitting reference: `sets_res/dcbg_table_v4_f005.json` (v4 report §5, pre-fix). D-CBG $\gamma = 4$ (§5.8): `run_l2r_dcbg.sh` (GAMMA=4.0) $\rightarrow$ `sets_res/dcbg_table_l2r_f005_g4.0.json` , `dcbg_table_l2r_f01_g4.0.json` , 2026-07-30, post-fix. Adj+D-CBG (§5.9): `run_l2r_dcbg.sh` (ADJP=1, GAMMA $\in$ {1.0, 4.0}) $\rightarrow$ `sets_res/dcbg_table_l2r_f005_adj.json` , `dcbg_table_l2r_f005_g4.0_adj.json` , 2026-07-30, legal-candidate-only enumeration. Three-seed replication (§5.10): `run_seed_stats.sh` $\rightarrow$ `sets_res/seed_stats_f005.json` , seeds 7/8/9, 2026-07-31. IW tempering (§8): `run_wg4.sh` $\rightarrow$ `sets_res/va_table_wg4_f005.json` , `wg4_ess.json` , 2026-07-31; math note `pdfs/temp_math.pdf` (src `report_src/temp_math.tex`). Reveal-order branches:

8. D-CBG vs IW, both at $\gamma = 4$ — unseen, family 0.05

The three mask-free guided arms with the guidance strength raised to $\gamma = 4$ on both sides, removing the " γ was swept for D-CBG only" asymmetry: D-CBG tilts the reveal posterior by $p_\phi(y|x)^4$ (\$5.8 runs), and IW raises its SNIS weights to $(D/(1-D))^4$, targeting $q^y p_\theta^{1-y}/Z$ — derivation, limits and the ESS-bias bound in pdfs/temp_math.pdf . No adjacency restriction anywhere. Setup otherwise unchanged: l2r, fam 0.05, unseen (e99 scorers $\rightarrow$ except_0 zero-shot), the 1,000 reserved pairs, RAW, seed 7, $n_{is} = 100$. Parentheses: gain over unseen base; bold: best of the three arms in the block.

blk	Experiments	valid	arrival	valid & arrival	EM	PC
1	D-CBG (first-order, $\gamma = 4$)	0.314 (+0.002)	0.997 (+0.001)	0.314 (+0.002)	0.275 (0.000)	0.290 (+0.001)
	D-CBG (exact, $\gamma = 4$)	0.393 (+0.081)	0.980 (−0.016)	0.387 (+0.075)	0.317 (+0.042)	0.340 (+0.051)
	IW-only ($w_\gamma = 4$)	0.339 (+0.027)	0.989 (−0.007)	0.335 (+0.023)	0.282 (+0.007)	0.302 (+0.013)
2	D-CBG (first-order, $\gamma = 4$)	0.268 (0.000)	0.963 (0.000)	0.262 (0.000)	0.245 (+0.001)	0.255 (+0.001)
	D-CBG (exact, $\gamma = 4$)	0.397 (+0.129)	0.947 (−0.016)	0.365 (+0.103)	0.315 (+0.071)	0.343 (+0.088)
	IW-only ($w_\gamma = 4$)	0.336 (+0.068)	0.962 (−0.001)	0.324 (+0.062)	0.303 (+0.059)	0.317 (+0.062)
4	D-CBG (first-order, $\gamma = 4$)	0.290 (−0.003)	0.976 (0.000)	0.282 (−0.003)	0.263 (+0.001)	0.275 (0.000)
	D-CBG (exact, $\gamma = 4$)	0.403 (+0.110)	0.945 (−0.031)	0.366 (+0.081)	0.316 (+0.054)	0.348 (+0.073)
	IW-only ($w_\gamma = 4$)	0.344 (+0.051)	0.974 (−0.002)	0.334 (+0.049)	0.306 (+0.044)	0.321 (+0.047)
8	D-CBG (first-order, $\gamma = 4$)	0.270 (−0.002)	0.963 (−0.001)	0.254 (−0.002)	0.230 (−0.002)	0.246 (−0.002)
	D-CBG (exact, $\gamma = 4$)	0.441 (+0.169)	0.910 (−0.054)	0.367 (+0.111)	0.316 (+0.084)	0.359 (+0.111)
	IW-only ($w_\gamma = 4$)	0.333 (+0.061)	0.961 (−0.003)	0.312 (+0.056)	0.296 (+0.064)	0.313 (+0.064)
16	D-CBG (first-order, $\gamma = 4$)	0.262 (+0.003)	0.932 (+0.002)	0.242 (+0.002)	0.224 (+0.003)	0.241 (+0.003)
	D-CBG (exact, $\gamma = 4$)	0.443 (+0.184)	0.867 (−0.063)	0.336 (+0.096)	0.297 (+0.076)	0.345 (+0.108)
	IW-only ($w_\gamma = 4$)	0.324 (+0.065)	0.927 (−0.003)	0.284 (+0.044)	0.265 (+0.044)	0.288 (+0.050)
32	D-CBG (first-order, $\gamma = 4$)	0.251 (0.000)	0.896 (−0.002)	0.221 (0.000)	0.202 (0.000)	0.221 (0.000)
	D-CBG (exact, $\gamma = 4$)	0.466 (+0.215)	0.823 (−0.075)	0.327 (+0.106)	0.297 (+0.095)	0.352 (+0.131)
	IW-only ($w_\gamma = 4$)	0.329 (+0.078)	0.897 (−0.001)	0.277 (+0.056)	0.264 (+0.062)	0.289 (+0.069)
64	D-CBG (first-order, $\gamma = 4$)	0.235 (0.000)	0.902 (+0.001)	0.202 (0.000)	0.178 (0.000)	0.200 (0.000)
	D-CBG (exact, $\gamma = 4$)	0.526 (+0.291)	0.768 (−0.133)	0.336 (+0.134)	0.279 (+0.101)	0.356 (+0.157)
	IW-only ($w_\gamma = 4$)	0.333 (+0.098)	0.869 (−0.032)	0.257 (+0.055)	0.269 (+0.091)	0.292 (+0.093)

- **At matched $\gamma = 4$, exact D-CBG leads IW-only at every block** — by +0.032 to +0.079 v_{Δ} — and the first-order variant stays inert (−0.003 to +0.002 vs base). The mechanism is asymmetric, not the formula: D-CBG's enumeration tilts the *whole vocabulary*, so $\gamma = 4$ can promote tokens p_θ would rarely propose; the IW tilt can only reweight the $n_{is} = 100$ candidates the proposal already drew, and raising w_γ $1 \rightarrow 4$ accordingly buys only +0.001 to +0.011 v_{Δ} (measured ESS stays at 94.6–99.6 / 100) — exactly the $\text{Var}(\ell)$ -limited movement temp_math.pdf Prop. 3(iii) predicts for a proposal-restricted candidate pool.
- **Read next to \$5.9/\$5.10 before concluding.** This is the one panel where the published baseline wins, and it is also the panel where all three arms are far below the masked arms (best cell here 0.387 v_{Δ} vs 0.574 with the mask). Once the Lemma-3 restriction is on, the ordering reverses on EM/PC (Adj+IW leads, robust over three seeds) and the v_{Δ} gap closes. The honest summary: *enumeration + tempering wins the unmasked regime; masking + lookahead wins the masked one, at a twelfth of D-CBG-exact's unmasked cost.*
- **The productive axis for the IW side is the candidate pool, not the exponent:** ancestral within-block proposals (E6) and best-of-n composition (E1) widen what the weights can choose among; w_γ only sharpens choices already available.